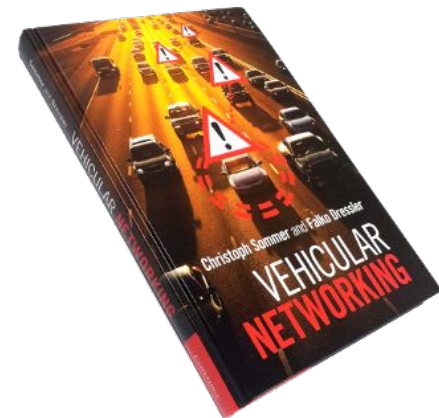


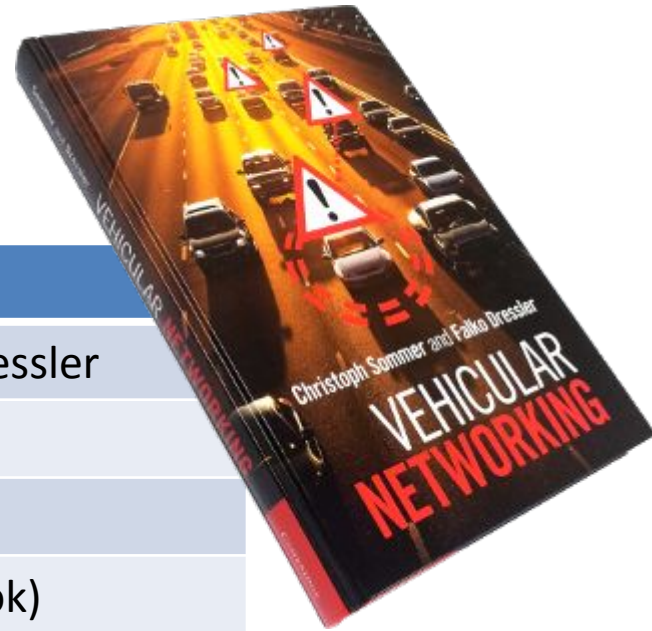
CSE 4615:: Wireless Networks

Understanding Vehicular Communication

Lecture 8.1

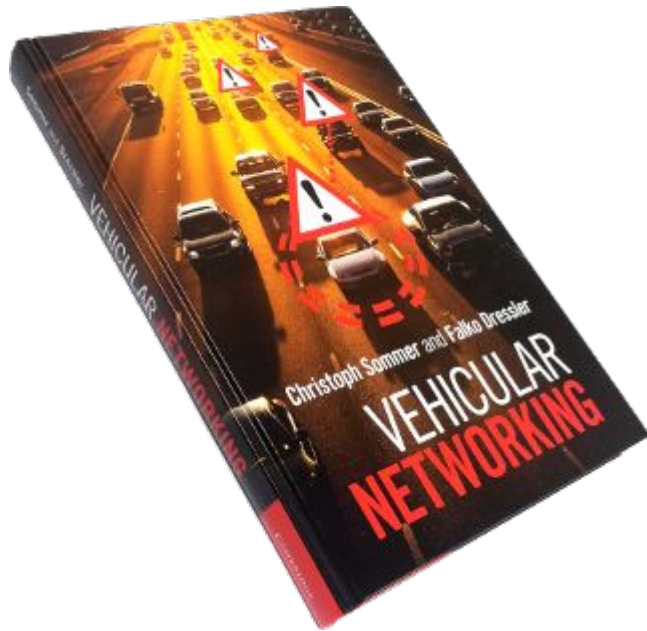


Course Material



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Website	http://book.car2x.org

Book design © 2014 Cambridge University Press



Vehicular Networking

Book design © 2014 Cambridge University Press

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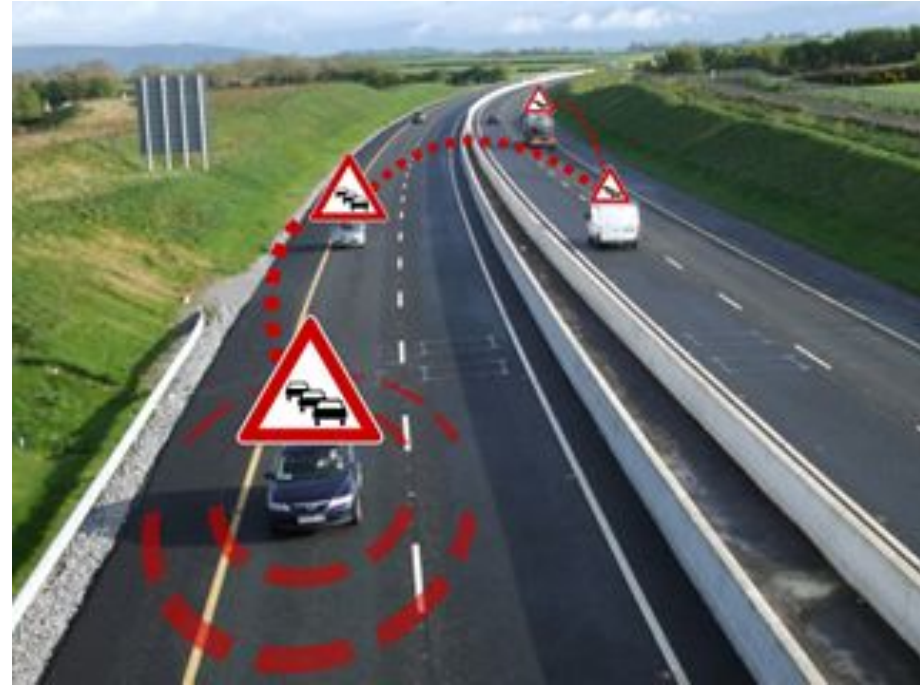
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A Short Introduction ...to Vehicular Networking



- Illustration © 2010 Christoph Sommer

Introduction of VANET

- **Vehicular Ad-Hoc Network** or **VANET** is an application of MANET that uses moving vehicles as nodes to create a mobile network.

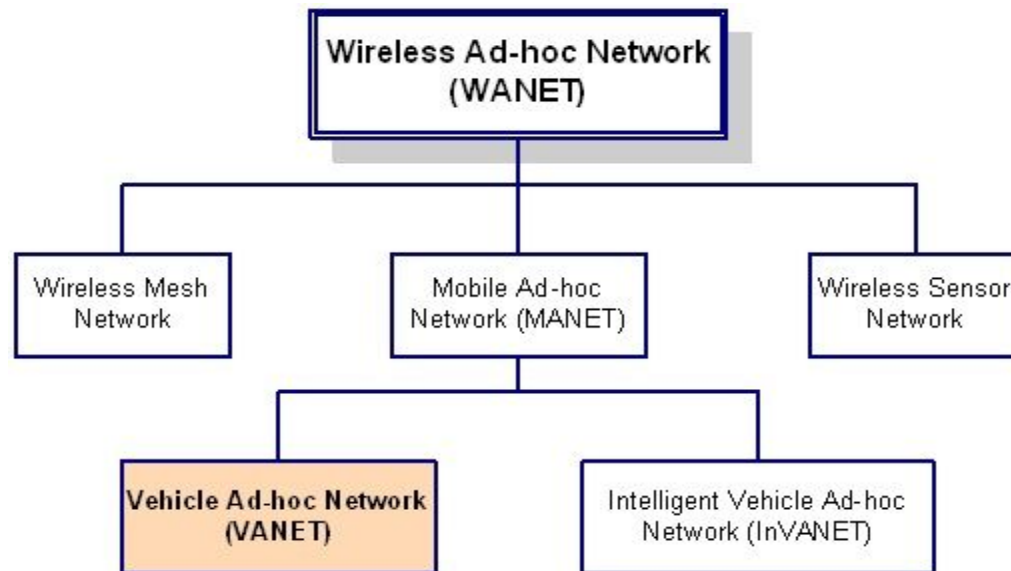


Fig. 1. Types of Networks

Cont...

- Vehicle-to-Vehicle Connectivity
 - Each participating vehicle acts as a wireless node/router.
 - Vehicles within approximately 300 m–1 km can communicate with each other.
 - Multiple connected vehicles form a large-scale network.
- Vehicle-to-Infrastructure Communication
 - Enables communication between vehicles and roadside infrastructure.
 - Helps make transportation systems smarter and more efficient.
 - Fixed infrastructure can be deployed at critical road locations.

Cont...

- Primary Goal: Road Safety
 - Shares important information such as:
 - Vehicle speed
 - Location/GPS coordinates
 - Traffic and road conditions
 - Communication can work with or without fixed infrastructure.
- Additional Services
 - VANET can also support value-added services such as:
 - Email and messaging
 - Audio/video sharing
 - Other infotainment services
- Example
 - Internal communication networks used by transport companies or travel agencies to connect and monitor their vehicles.

Components of VANET

- The major components of a VANET are :
 - On-Board Unit (OBU)
 - Road Side Unit (RSU)
 - Trusted Authority (TA)

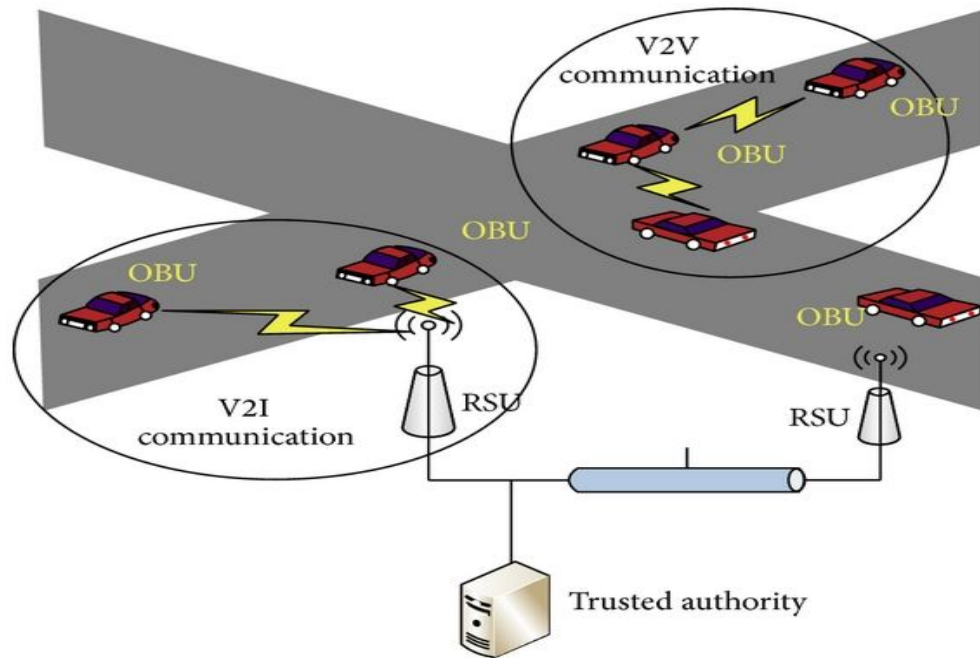


Fig. 2. VANET Architecture[4]

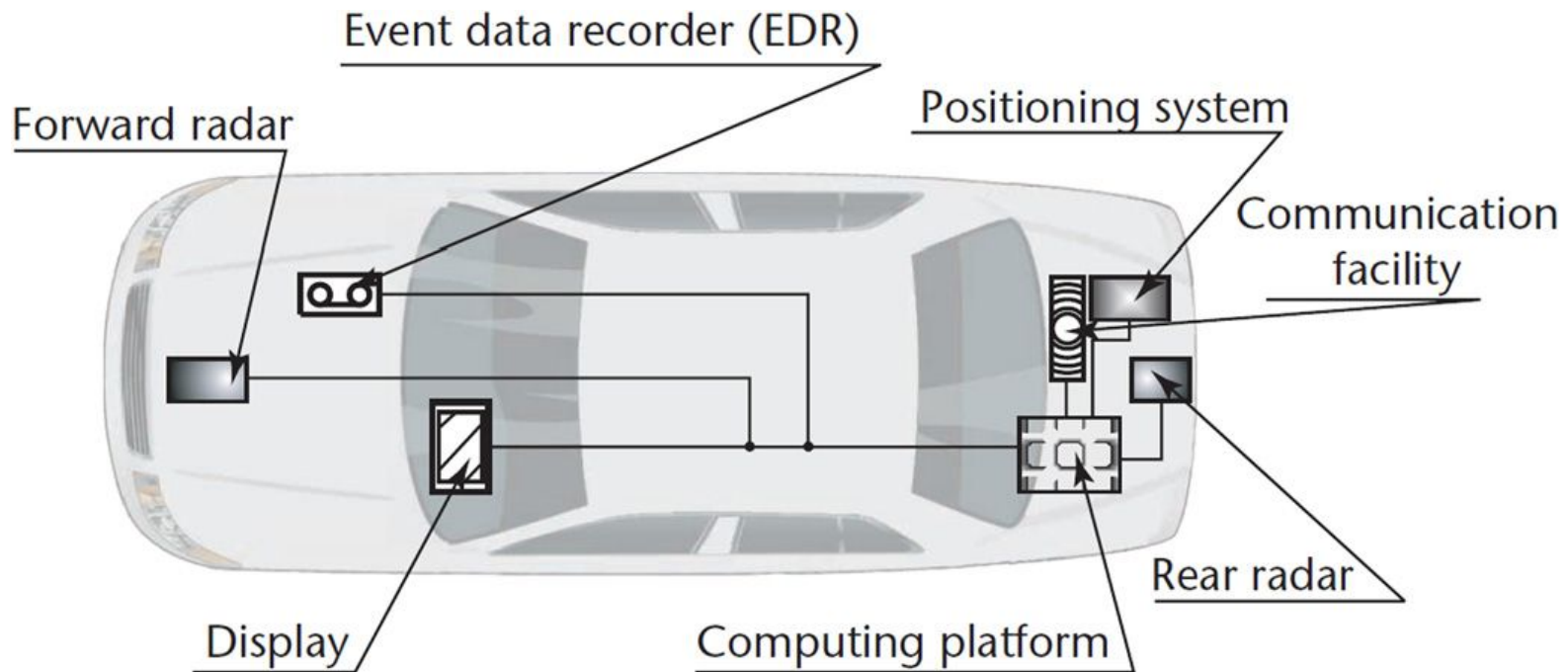
Cont...

- **OBU (On-Board Unit)**
 - Installed in vehicles to provide wireless communication.
 - Integrates display, and driver interaction
 - Enables vehicles to exchange information with other vehicles and infrastructure.
 - Acts as a key communication unit in Intelligent Transportation Systems (ITS)
- **RSU (Roadside Unit)**
 - Deployed at intersections and hotspots.
 - Provides information and Internet access to vehicles within its radio coverage.
- **TA (Trusted Authority)**
 - Installs secure parameters in the OBU.
 - Responsible for user authentication and security.

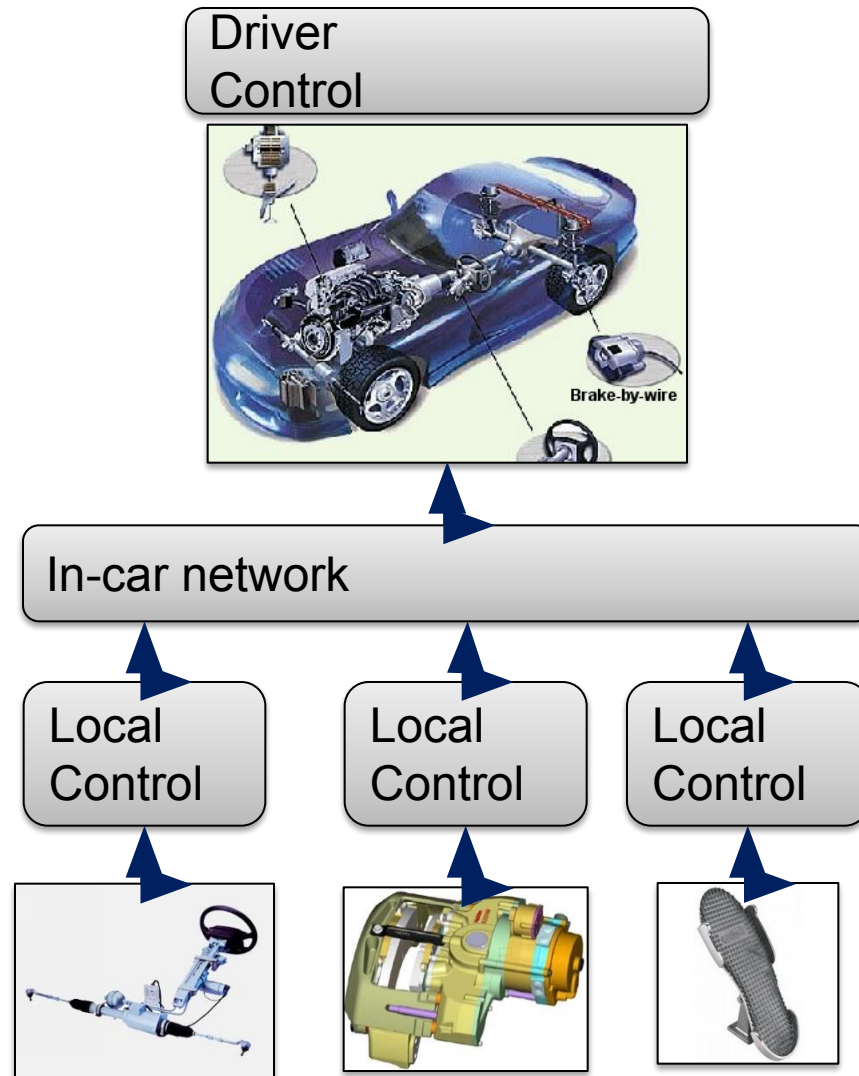
Cont...

- **VANET Communication Standards**
 - **DSRC** – Dedicated Short Range Communications
 - **WAVE** – Wireless Access in Vehicular Environments
 - **IEEE 802.11p** – Wireless communication standard for VANETs
 - These standards support the **VANET protocol stack**
- **Intelligent Vehicle Components**
 - **Sensors:** Front/reversing radar for environmental information
 - **GPS:** Vehicle positioning and driving assistance
 - **Communication System:** Enables vehicle-to-vehicle and infrastructure communication
 - **Computing System:** Processes collected information
 - **Event Recorder:** Records events similar to an aircraft black box
- **Vehicle Security & Identification**
 - **ELP (Electronic License Plate)** – Electronic vehicle identification
 - **ECN (Electronic Chassis Number)** – Digital identity of the vehicle

Smart Vehicle

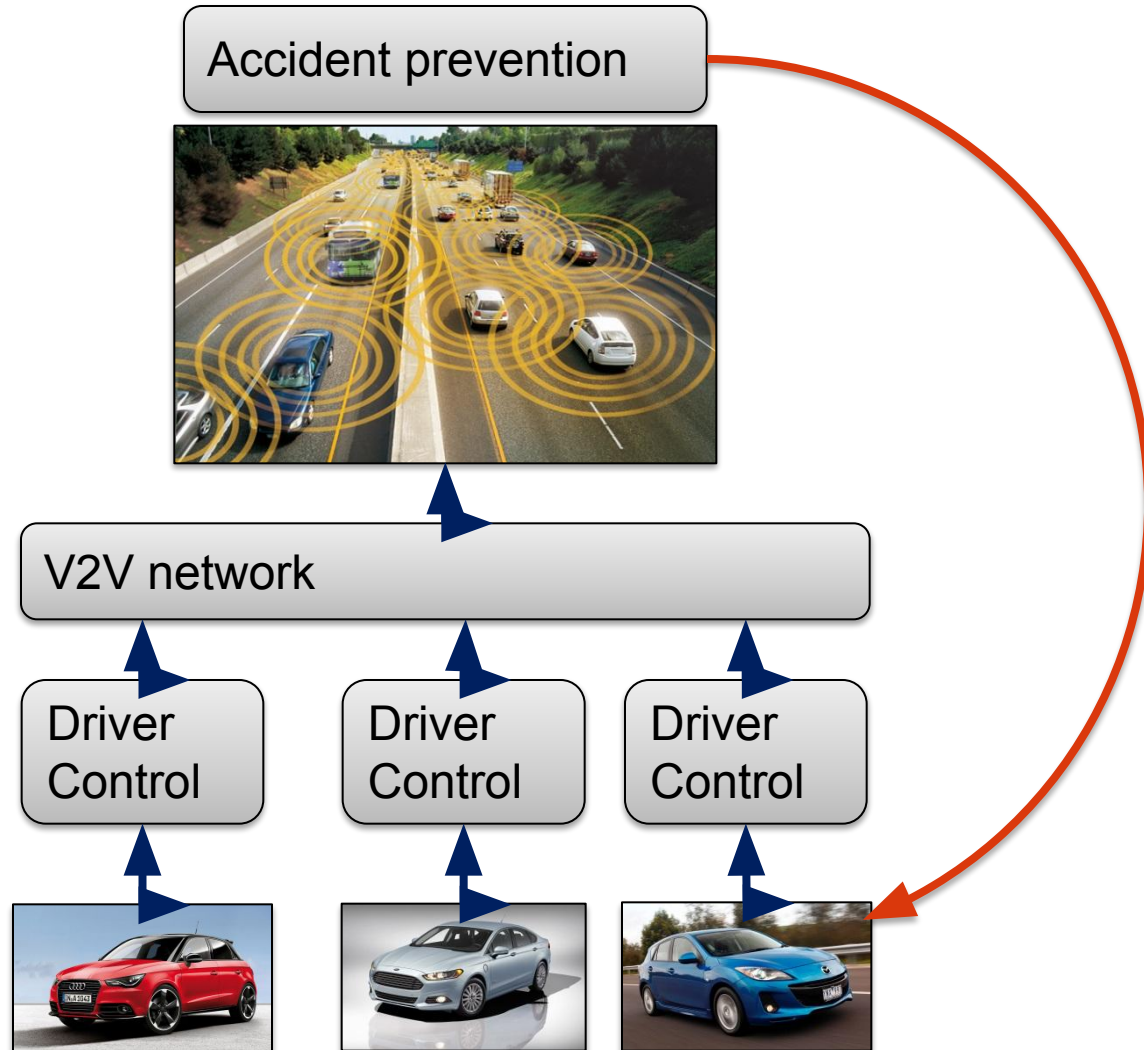


Vehicles operate using networked ICT



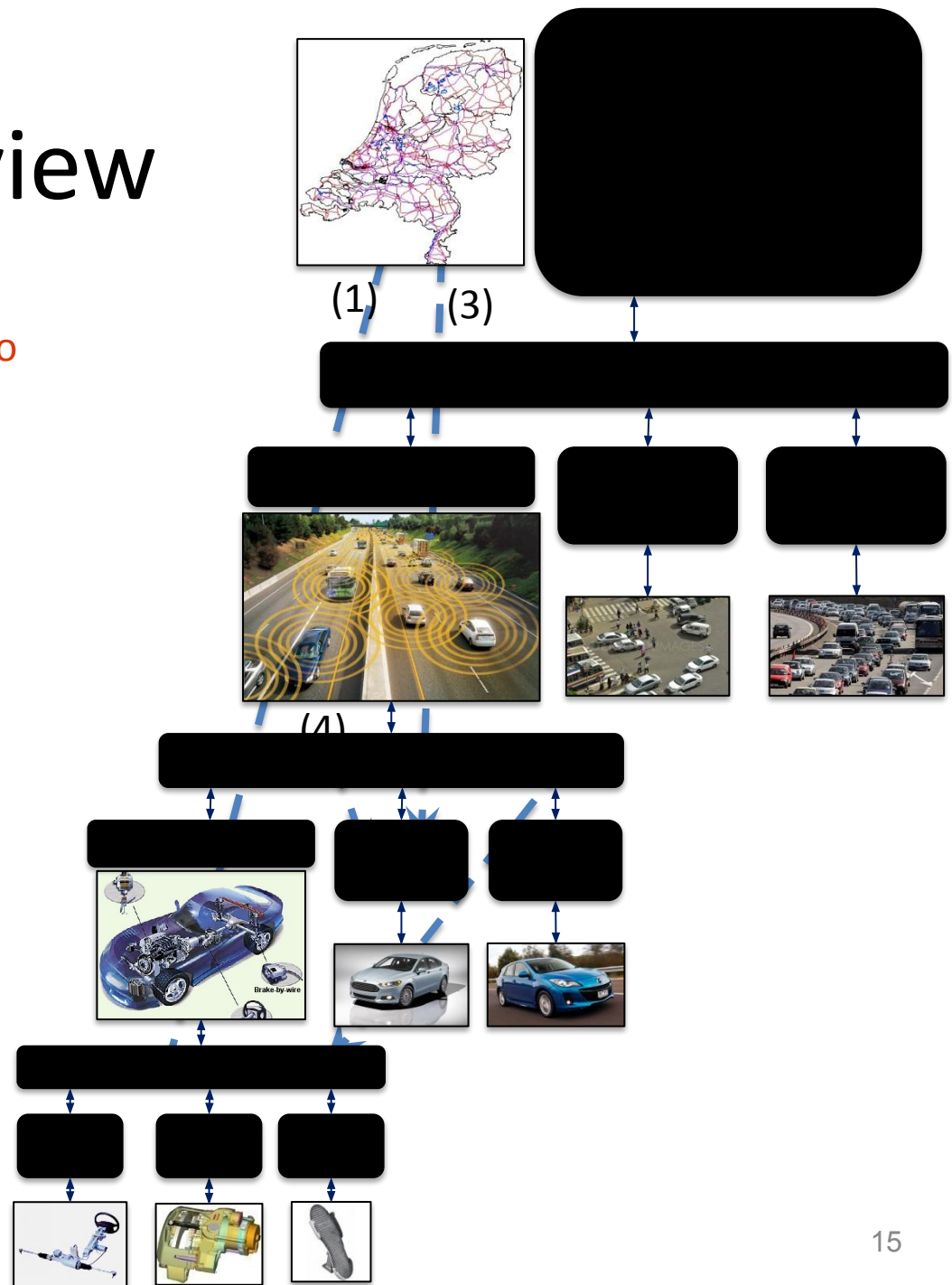
Vehicles become parts of a larger whole

- Vehicle to Vehicle (V2V) communication
 - ad-hoc
 - collaborative applications
- Applications are distributed
- Actuation may pass by driver control

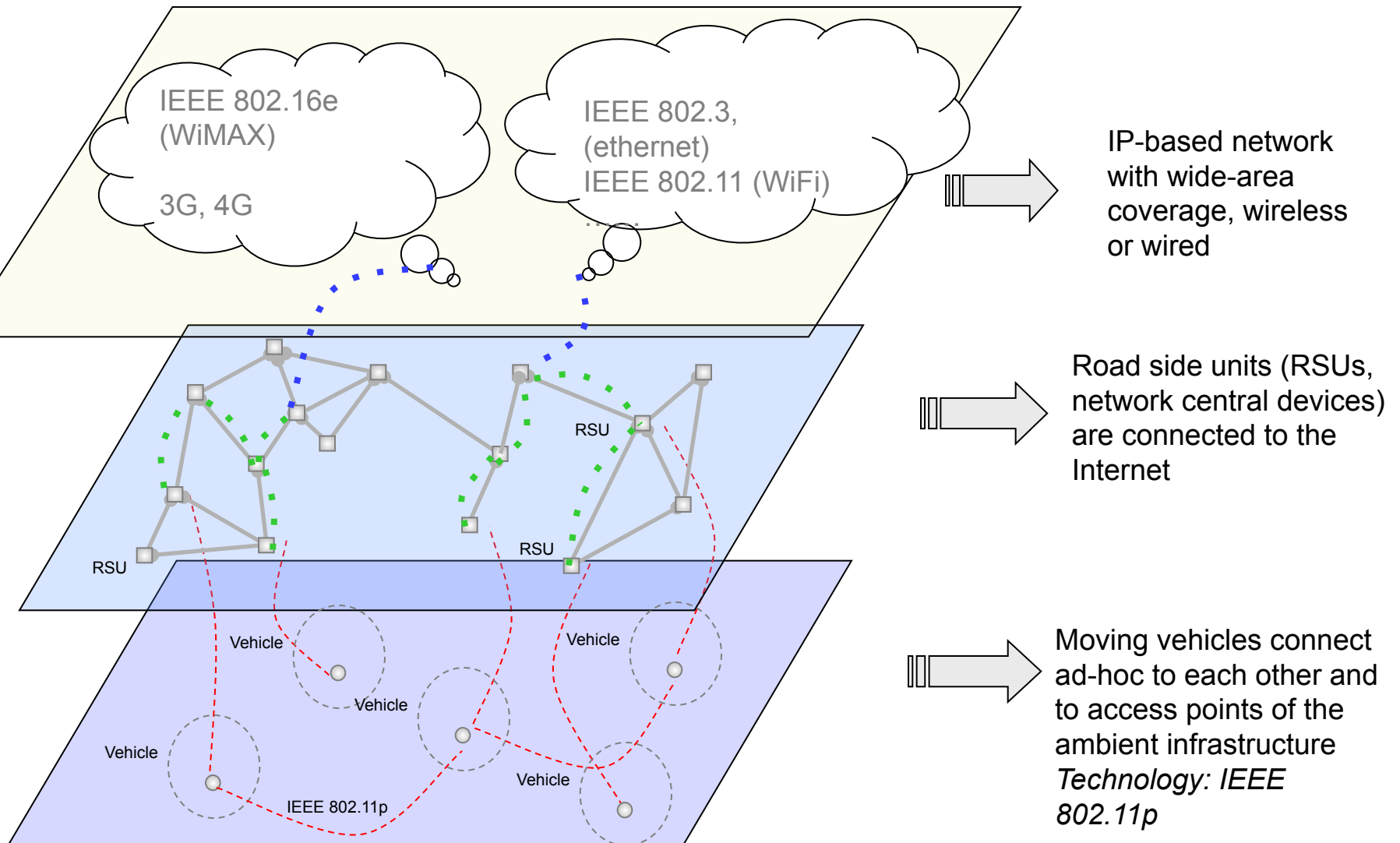


A conceptual view

- Example data flows:
 - (1) gather detailed driving data to determine
 - local weather
 - road condition
 - (2) accident prevention by direct intervention
 - (3),(4) informing driver about upcoming road conditions



Physical organization in V2V



Executive Summary



The Concept	• vehicle-to-vehicle (V2V).
The Opportunity	• Smart Cars
The Potentials	• Safer roads • Less traffic • Less fading

Visionary Applications

- Lane assistant
 - Simple roadside beacons support lane detection
- Lateral collision avoidance
 - More advanced beacons on cars and motorcycles help maintain minimum separation
- Accident reporting
 - Broken down cars can automatically send simple report to central server
- Intersection assistance
 - Pairs of cars automatically coordinate complex maneuvers at intersections
- Cooperative driving
 - The future evolution of autonomous driving: vehicles actively support each other's route planning, navigating, driving

Visionary Applications

- ...and much (much) more:
- Emergency Brake Light Warning,
- Accident Warning,
- Emergency Flashlights,
- Traffic Jam Warning,
- Weather Warning,
- Emergency Vehicle,
- Slow Vehicle,
- Moving and Static Road Works,
- Obstacle Warning,
- Intersection Maneuvering Assistance,
- Intersection Traffic Lights,
- Lane Change,
- Maneuvering Assistance,
- Longitudinal Maneuvering Assistance,
- Floating Car Data Collection,
- Free-Flow Tolling,
- Breakdown Call,
- Remote Diagnostics,
- Theft Detection,
- Emergency Call,
- Just-In-Time Repair Notification,
- Emergency Traffic Light Pre-emption,
- Traffic Light Assistant,
- Interactive Lights Dimming,
- Emergency Traffic Light Pre-emption,
- Traffic Light Assistant,
- Roadside Traffic Camera pull,
- In-vehicle signing pull,
- Regional Information pull,
- Electronic Payment pull,
- Logistic for goods being loaded and unloaded,
- Traffic Information Service pull,
- Traffic Information Service push,
- In-vehicle signing push,
- Telemetric Onboard/Off-board Diagnostics,
- Remote Vehicle Status Control,
- Fleet Management,
- Server based navigation,
- Remote lock-down,
- Remote entry,
- Mobile Office,
- General internet services,
- Internet Audio,
- Web Browsing,
- Electronic Toll Collection,
- Parking Unit Fee Payment,
- Goods and services discovery and payment,
- Guided Tour, ...

Visionary Applications (Classified)

- The aim of VANET is to improve driving experience and the safety of transportation.
- 1. **Safety Application (to enhance driving safety):**
 - EEBL – Emergency Electronic Brake Light (sudden braking)
 - PCN – Post Crash Notification
 - RFN – Road Feature Notification (ex. downhill curve)
 - LCA – Lane Change Assistance

2. Convenience Application (for better driving experience):

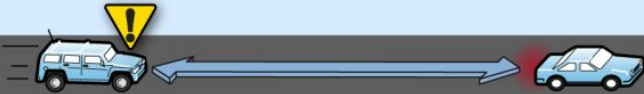
- Road Congestion Notification
- Dynamic Route/Travel Time Planning
- Finding Parking Spots

3. Other Commercial/ Infotainment Application:

- Free flow Tolling
 - Social Networking
 - Multimedia Content Exchange
-

C'est le Rush

<https://www.youtube.com/watch?v=Johmxw3cspA>

Scenario and warning type	Scenario example
Rear end collision scenarios	Forward collision warning Approaching a vehicle that is decelerating or stopped. 
	Emergency electronic brake light warning Approaching a vehicle stopped in roadway but not visible due to obstructions. 
Lane change scenarios	Blind spot warning Beginning lane departure that could encroach on the travel lane of another vehicle traveling in the same direction; can detect vehicles not yet in blind spot. 
	Do not pass warning Encroaching onto the travel lane of another vehicle traveling in opposite direction; can detect moving vehicles not yet in blind spot. 
Intersection scenario	Blind intersection warning Encroaching onto the travel lane of another vehicle with whom driver is crossing paths at a blind intersection or an intersection without a traffic signal. 

Source: GAO analysis of Crash Avoidance Metrics Partnership information.

from: Vehicle-to-Vehicle Communications: Readiness of V2V Technology for Applications, NHTSA, August 2014

Some application examples

(US WAVE BSM ~SAE J2735)

Apps.	Comm.type	Freq.	Latency	Range
Lane Change Warning	V2V, periodic, P2M	10Hz	100ms	150m
Collision Warning	V2V, periodic, P2M	10Hz	100ms	150m
Emergency Brake Lights	V2V, event-driven, P2M	10Hz	100ms	300m
Pre-Crash Sensing	V2V, event-driven, P2P	50Hz	20ms	50m
Stop Sign Assists	I2V and V2I, periodic	10Hz	100ms	250m
Left Turn Assistance	I2V and V2I, periodic, P2M	10Hz	100ms	300m
Traffic Signal Violation	I2V, periodic, P2M	10Hz	100ms	250m
Curve Speed Warning	I2V, periodic, P2M	1Hz	1s	200m

V2V = Vehicle to Vehicle
P2M = Point to Multipoint
I2V = Infra structure to Vehicle

Eight high priority vehicle safety applications as chosen by NHTSA and VSCC.

NHTSA – US National Highway Traffic Safety Administration

VSCC – Vehicle Safety Communication Consortium of CAMP (Crash Avoidance Metrics Partnership)

Johan J. Lukkien, j.j.lukkien@tue.nl
TU/e Informatica, System Architecture and
Networking

Cont...

- **Emergency Electronic Brake Light (EEBL)**
 - Purpose of EEBL
 - Broadcasts emergency/hard braking events to nearby vehicles.
 - Receiving vehicles assess the relevance of the warning.
 - Alerts the driver to help avoid potential crashes.
 - Especially useful when visibility is limited.
 - How EEBL Works
 - When a vehicle brakes hard, EEBL sends a Basic Safety Message (BSM) to surrounding vehicles.
 - Following vehicles receive an early warning of sudden braking.
 - Helps overcome visibility problems caused by:
 - Large vehicles/trucks
 - Fog
 - Heavy rain
 - Advantages Over Conventional Brake Lights
 - Extends the warning beyond direct line of sight.
 - Can provide additional information such as:
 - Deceleration rate
 - Duration of braking
 - Conventional brake lights only indicate that the brake is applied.
 - EEBL can provide a more informative and timely warning to following vehicles.

Cont...

- **Post-Collision Notification (PCN)**
 - Detects a collision/incident based on:
 - Vehicle speed
 - Deceleration rate
 - Helps identify and report post-crash events.
- **Lane Change Assist (LCA)**
 - Uses radar sensors to monitor:
 - Blind spots
 - Traffic behind the vehicle
 - Measures the closing speed of vehicles in adjacent lanes.
 - Determines whether a lane change is safe.
 - Provides audible and visual warnings when a lane change is unsafe.
- **Cooperative Collision Warning (CCW)**
 - Provides active safety for vehicles, especially on highways.
 - Exchanges static and dynamic vehicle information through wireless communication.
 - Calculates the relative safety distance between vehicles.
 - Stores received information in the Motor Vehicle Event Data Recorder (MVEDR).
 - Supports future accident investigation.

Cont...

- **Cooperative Collision Warning (CCW)**

- CCW Concept

- Provides collision warnings or situation awareness to drivers.
 - Uses wireless communication to obtain information from neighboring vehicles.
 - Does not require ranging sensors to detect nearby vehicles.
 - Monitors the location and movement of surrounding vehicles.

- Traditional Collision Warning Systems

- Use sensors such as:

- Radar / Lidar
 - Laser
 - Cameras/Vision

- Sensors monitor vehicles in:

- Front
 - Rear
 - Left lane
 - Right lane

- Wireless-Based CCW

- Builds awareness of the vehicle surroundings by receiving wireless messages from nearby vehicles.
 - Vehicles also transmit their own information to neighboring vehicles.
 - Enables cooperative and real-time collision awareness without relying solely on physical sensors.

Characteristics

- High Mobility
- Rapid Changing Network Topology
- Predictable Mobility
- No Power Criteria
- Time Management

Cont...

VANET is an application of MANET but it has its own distinct characteristics, the characteristics of VANETs are basically a mixture of wireless medium characteristics. The characteristics are:

1. **High Mobility:** - This is important features of the VANET as nodes move in a high speed all the time with different direction . The high mobility of nodes reduces the mesh in the network (fewer routes between nodes). Compared to MANET, VANET mobility is relatively high.
2. **Rapid Changing Network Topology:** - As nodes move in very high speed so the position of node changes frequently so therefore network topology in VANETs tends to change frequently. The connection times are short especially between nodes moving in opposite direction.
3. **No Power criteria:** - The VANET node is equipped with a battery that is used as an infinite power supply for the communication and computation task
4. **Time Management:** - Safety message are the main goal of VANET. Message in VANET must be delivered to the nodes within the time limit so that a decision can be made by the node and perform action according.
5. **Wireless Communication:** - Data transmission is generally done by nodes. Nodes are connected and exchange their information via wireless communication

Challenges of communication

- Basic challenges
 - Timeliness
 - Throughput
- Communication in vehicles: stresses...
 - Robustness
 - Cost
- Communication across vehicles: also needs...
 - Interoperability
 - Reachability
 - Security
 - Privacy

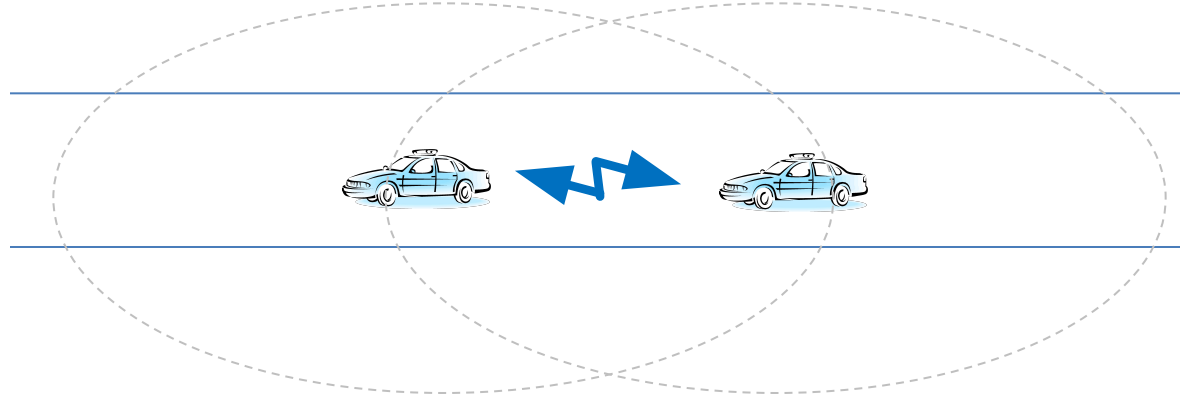
Cont...

- The main challenges of the VANETs can be summarized as follows:
 - Due to high mobility neighborhood location changes frequently-network Management.
 - The unbounded network size.(channel load) Congestion and collision Control:
 - Due to mobility signal power changes continuously.
 - Environmental impact VANETs uses the electromagnetic waves for communication.
 - Security as VANET provides the road safety applications which are life critical therefore security of these messages must be satisfied.

V2V: goals and characteristics

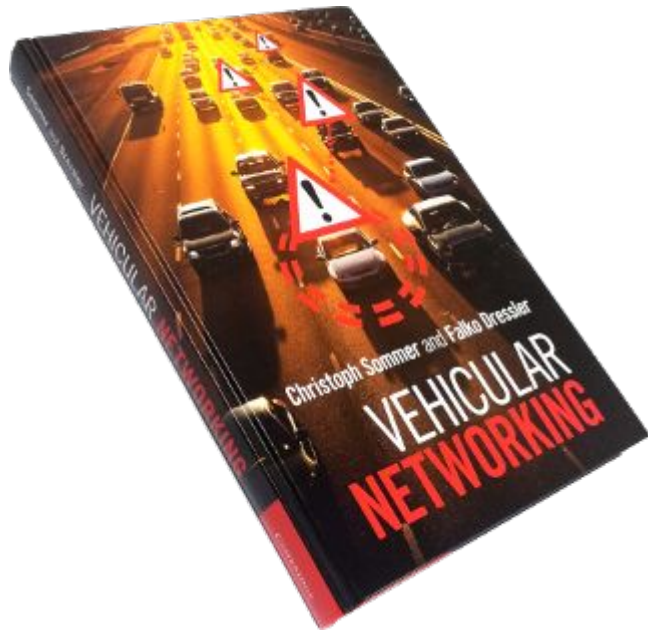
- Goals

- Safety
- Comfort
- Traffic efficiency



- Safety applications have strict requirements on

- Reliability
- Delay



Car-to-X Networking

Vehicle-to-X (V2X) communication patterns

- Vehicle-to-X (V2X),
- Inter-Vehicle Communication (IVC),
- Vehicular ad-hoc network (VANET),
- ...

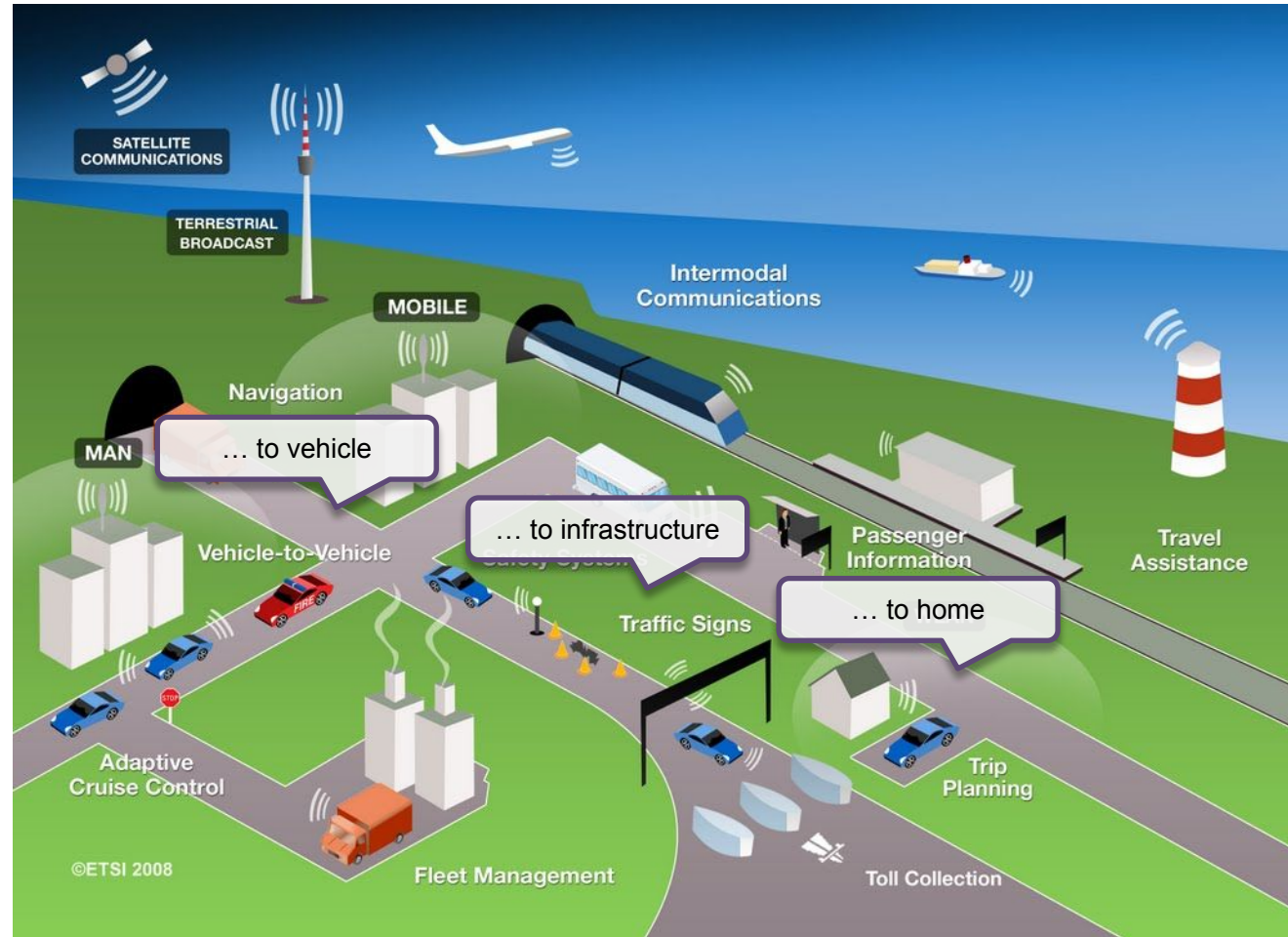


Illustration: ETSI

Taxonomy of Use Cases

Vehicle-to-X

Non-Safety

Safety

Comfort

Traffic
Information
Systems

Situation
Awareness

Warning
Messages

Contextual
Information

Entertainment

Optimal
Speed
Advisory

Congestion,
Accident
Information

Adaptive
Cruise
Control

Blind Spot
Warning

Traffic Light
Violation

Electronic
Brake Light

Taxonomy of Use Cases

Vehicle-to-X

Non-Safety

Many messages

High data rate

Low latency demands

Low reliability demands

Safety

Few messages

Small packet size

High latency demands

High reliability demands

Diversity of use cases

Application	Distance	Time	Recipient
Hazard warning	250m	10s	All
Location based service	1..5km	Weeks	Subscribers
City wide alarm	20km	Hours	All
Travel time information	5km	Minutes	All
File sharing	250m	Minutes (Index) Days (Content)	Subscribers (Index) Peers (Content)
Interactive Services	1..5km	Minutes	Subscribers

[1] Bai, F. and Krishnamachari, B., "Exploiting the Wisdom of the Crowd: Localized, Distributed Information-Centric VANETs," IEEE Communications Magazine, vol. 48 (5), pp. 138-146, May 2010

Diversity of requirements

Application	Latency	Reliability	# Vehicles	Area	Persistence
Information Query	★	★	★★★	★★★	
Hazard Warning	★★★	★★	★★	★★★	
ACC, el. Brake Light	★★★	★★	★	★	
Cooperative Awareness	★★	★★★★	★	★	★
Intersection Assistance	★★	★★★★	★★	★★	★
Platooning	★★★	★★★★	★★	★	★

[1] T. L. Willke, P. Tientrakool, and N. F. Maxemchuk, "A Survey of Inter-Vehicle Communication Protocols and Their Applications," IEEE Communications Surveys and Tutorials, vol. 11 (2), pp. 3-20, 2009

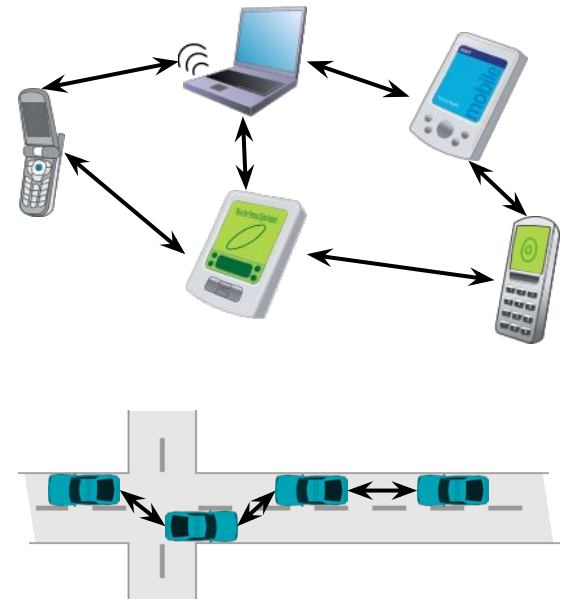
Motivation

- 1970s: bold ideas
 - Very visionary, infrastructure-less solutions
 - Unsupported by current technology
- Early interest of government and industry
 - working prototypes in: Japan *CACS* (1973–1979), Europe *Prometheus* (1986–1995), U.S. *PATH* (1986–1992)
 - No commercial success
- 1980s: paradigm shift
 - From complete highway automation ⇨ driver-advisory only
 - Infrastructure-less ⇨ infrastructure-assisted
 - *chicken-and-egg* type of standoff
- New technology re-ignites interest
 - latest-generation cellular communication ⇨ early “Car-to-X” systems
 - e.g., *On Star* (1995), *BMW Assist* (1999), *FleetBoard* (2000), and *TomTom HD Traffic* (2007).
- Sharp increase in computing power
 - Supports fully-distributed, highly reactive ad hoc systems

[1] W. Zimdahl, “Guidelines and some developments for a new modular driver information system,” in 34th IEEE Vehicular Technology Conference (VTC1984), vol. 34., Pittsburgh, PA: IEEE, May 1984, pp. 178–182.

Motivation

- Traditional Network
 - Connection: wired
 - Nodes: non-moving
 - Configuration: static
- Mobile Ad Hoc Network (MANET)
 - Connection: wireless
 - Nodes: mobile
 - Configuration: dynamic
 - (Infrastructure: optional)
- Vehicular Ad Hoc Network (VANET)
 - Not: “MANET on wheels”
 - Different topology dynamics, communication patterns, infrastructure, ...



[1] M. Scott Corson and Joseph Macker, "Mobile Ad hoc Networking (MANET): Routing Protocol Performance Issues and Evaluation Considerations," RFC 2501, January 1999

Freeway



Urban

- 1D mobility
- Bimodal connectivity
 - Stable connection
(vehicles on same lane)
 - AND
 - unstable connection
(vehicles on opposite lane)
- High speed
- ...

- 2D mobility
- Bipolar connectivity
 - Many neighbors
(when standing)
 - OR
 - Few neighbors
(when driving)
- Obstacles
- ...

Levels of infrastructure support

- Pure ad hoc communication
- Stationary Support Units (SSU)
 - Radio-equipped autonomous computer
 - Inexhaustible storage, energy supply
 - Known position, high reliability
- Roadside Units (RSU)
 - SSU(System Synchronization Unit) plus...
 - Ethernet NIC, UMTS radio(3G Universal Mobile Telecommunications System), ...
 - Connected to other RSUs
- Traffic Information Center (TIC)
 - Central server connected to RSUs

Infrastructure



No Infrastructure

- Central coordination
 - Resource management
 - Security
 - High load on core network
- Self organizing system
 - Channel access
 - Authentication
 - Low data rate

Source: AKTIV CoCar

Convergence towards heterogeneous approaches

- Same system needs to work in multiple environments
 - Vehicle starts to drive in city with infrastructure support
 - Continues driving on freeway (still with infrastructure support)
 - Loses infrastructure support when turning onto local highway
 - Finishes driving in city without infrastructure support

Challenges of V2X communication

Communication

- Highly varying channel conditions
- High congestion, contention, interference
- Tightly limited channel capacity

Networking

- Unidirectional Links (Broadcast, No Response from Receiver)
- Multi-Radio / Multi-Network
- Heterogeneous equipment

Mobility

- Highly dynamic topology
- But: predictable mobility
- Heterogeneous environment

Security

- No (or no reliable) uplink to central infrastructure
- Ensuring privacy
- Heterogeneous user base

Technology

Communication paradigms and media

Wireless Communication Technologies

Infrastructure-based

Infrastructureless

Broadcast

Cellular

Short Range

Medium Range

FM Radio,
DAB/DVB,
...

GSM
2G
Cellular

UMTS
3G
Cellular

LTE /
WiMAX
4G Cell.

Millimeter,
Infrared,
Visible

802.15.1
Bluetooth

802.15.4
ZigBee

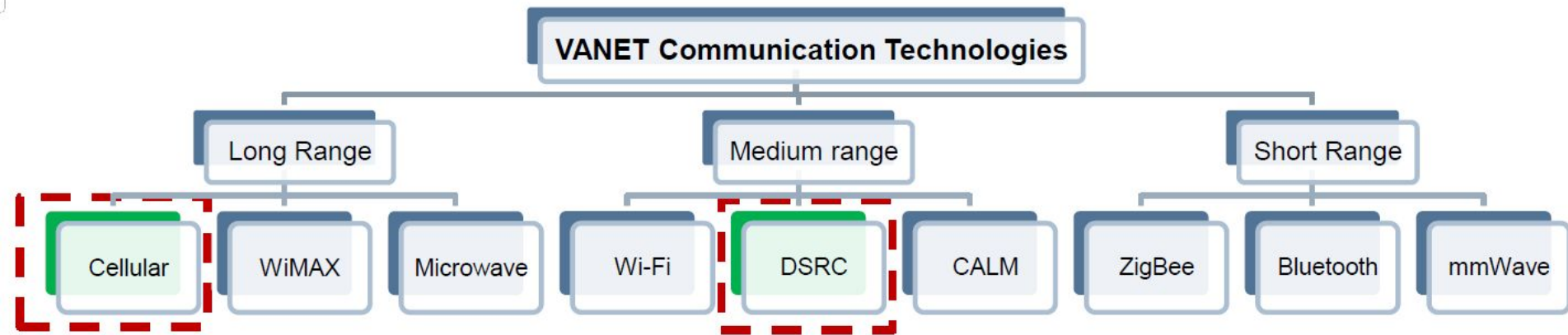
802.11
Wi-Fi

DSRC /
WAVE

[1] Dar, K. et al., "Wireless Communication Technologies for ITS Applications," IEEE Communications Magazine, vol. 48 (5), pp. 156-162, May 2010

Communication paradigms and media

Cellular and **DSRC** prove to be potential communication technologies enabling connected cars



The two major vehicular wireless communication technological standards:

1. IEEE 802.11p (DSRC/WAVE)
2. Cellular V2X (C-V2X)

Communication Technologies

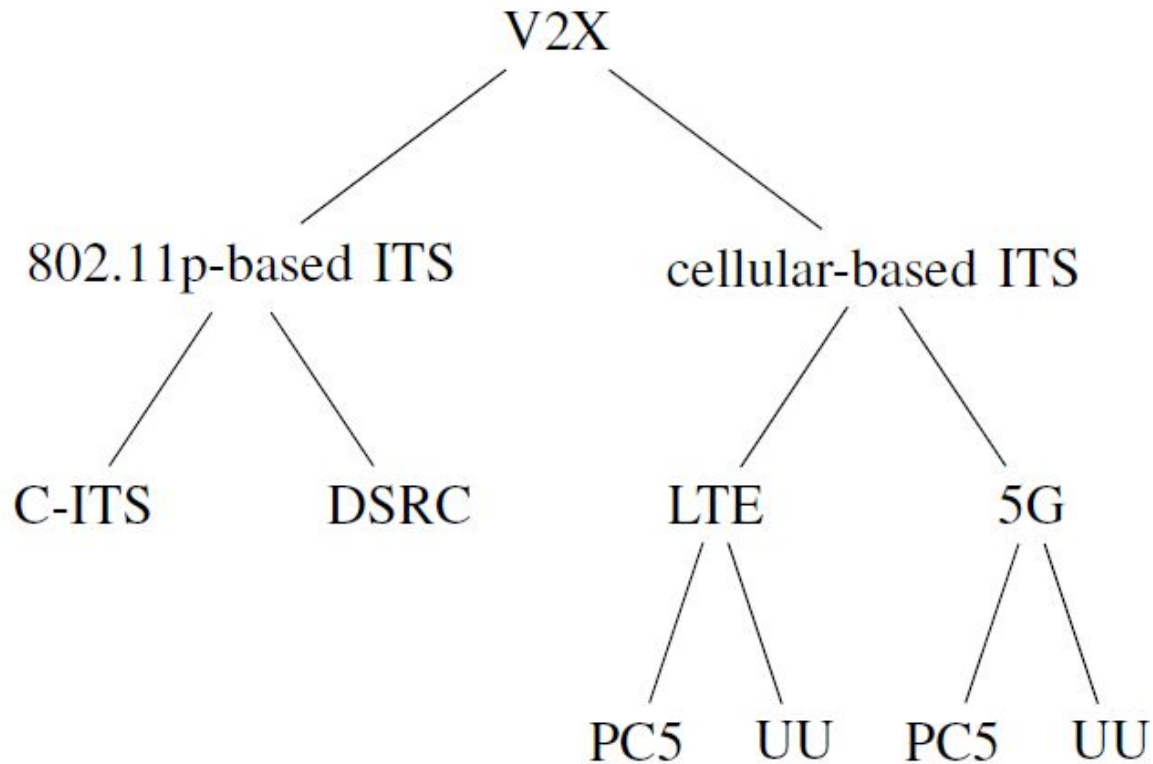
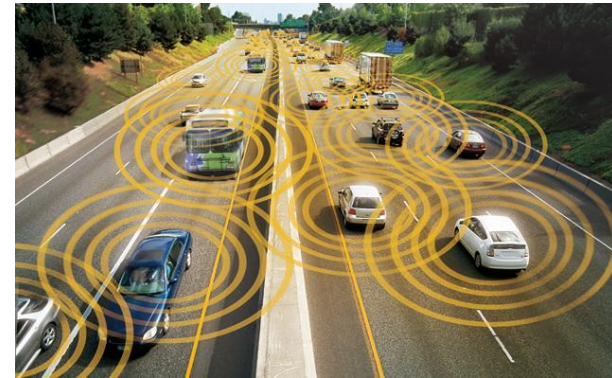


Figure 1: Taxonomy of used V2X communications

How does this work?

- It is *cooperative, dynamic* and *ad-hoc*
- Two different approaches, same network technology (IEEE 802.11p)
 - **US:** *Wireless Access in Vehicular Environments* – WAVE – over Dedicated Short Range Communications – DSRC
 - **EU:** ETSI TC ITS standards, using Geo-networking
- Essentially: vehicles emit *periodically* or *event-driven* status information
 - called *Basic Safety Messages* (BSM, US)
 - and *Cooperative Awareness Messages* (CAM, EU)



picture from the Internet

IEEE 802.11p characteristics

- Part of the full IEEE 802.11 (2800page) specification
 - enhancements for use in Vehicles
- **Connect without BSS** (BSS: stations with access point):
 - no association, authentication
 - just direct, ad-hoc messaging when in range
- Support prioritizing of traffic: EDCA (from 802.11e)
- Deal with high relative speeds of terminals
- UTC-based timing reference for synchronizing time accurately
- Channels in the 5.9 GHz range (~300m in free field, max 1000)

US: WAVE/DSRC

- IEEE 802.11p standard

- A Dedicated Short Range Communication (DSRC) is proposed which is operating on 5.9GHz band and uses 802.11 access methods
- provides short range communication with low latency.
- add wireless access in vehicular environments (V2V, I2V Communications)

- IEEE 1609 standard

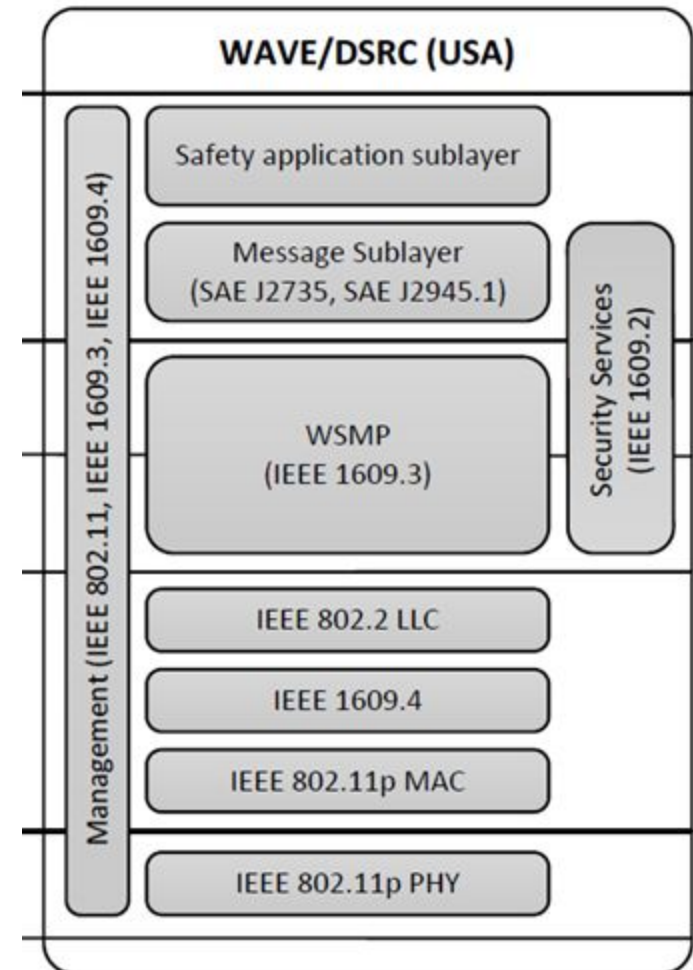
- family of standards that function in the middle layers of the protocol stack to flexibly support safety applications in VANETs .

- J2735 standard

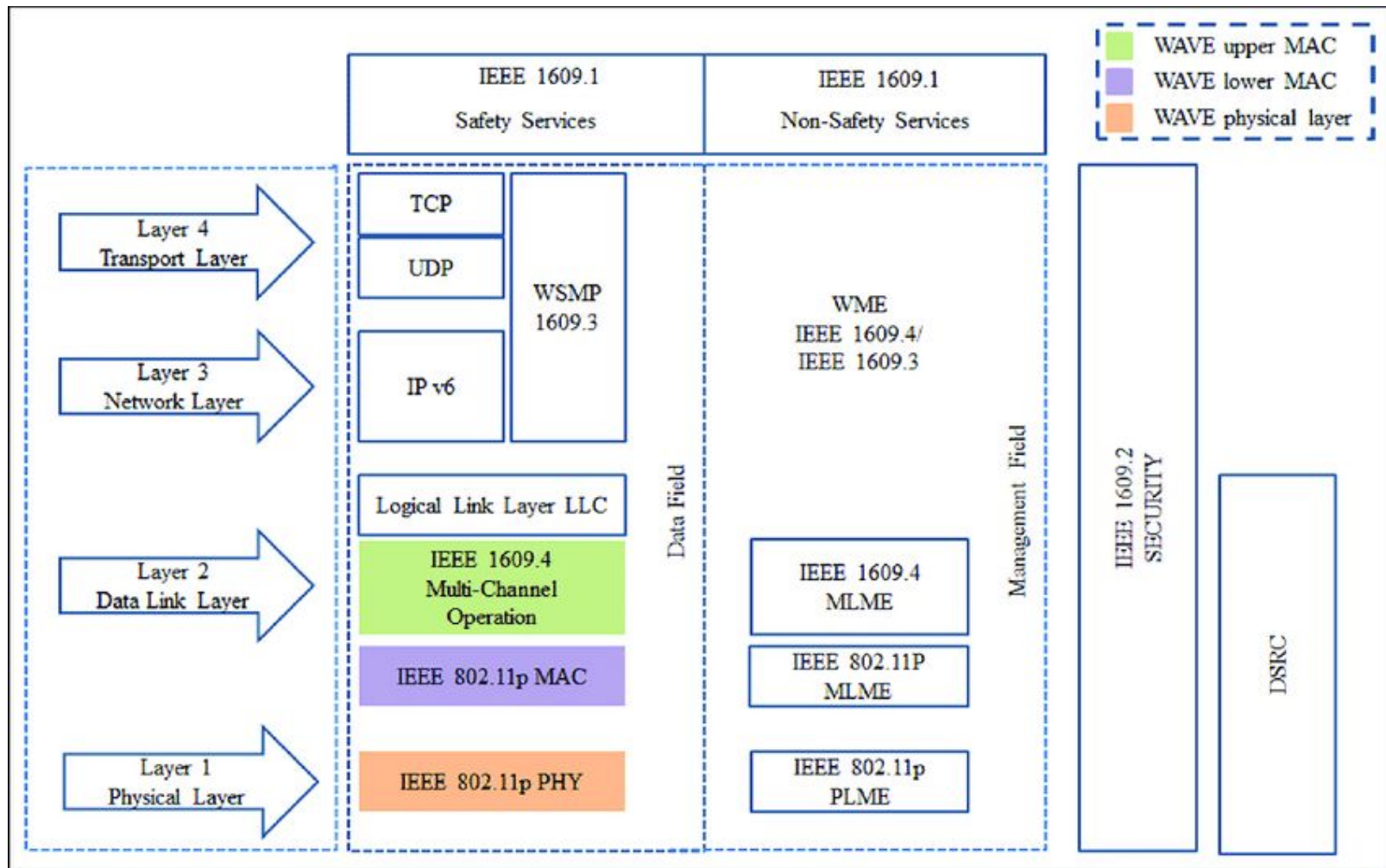
- placed in the application layer
- It defines message sets, data frames and elements which are used for V2V and V2I safety exchanges

US: WAVE/DSRC

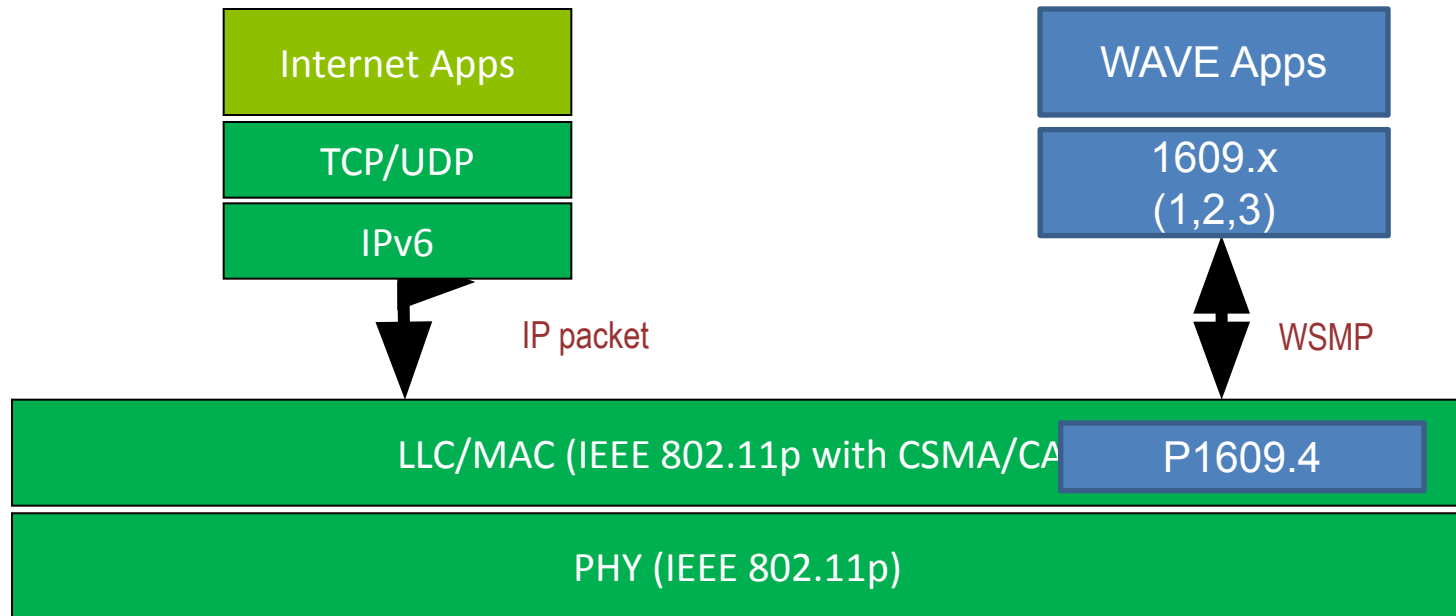
- IEEE 1609 standardizes *Wireless Access in Vehicular Environments over Dedicated Short Range Communication*
 - 1609.1: resource management
 - 1609.2: security services
 - 1609.3: networking services (a.o. wave short message protocol)
 - broadcasting of basic safety messages
 - define channel number and power per message
 - 1609.4; multi-channel operation
- SAE J2735
 - defines the message content of WSMP



Protocol Stack: IEEE 802.11p (DSRC/WAVE)



Combining with Internet



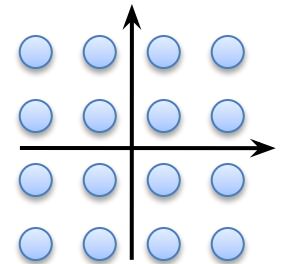
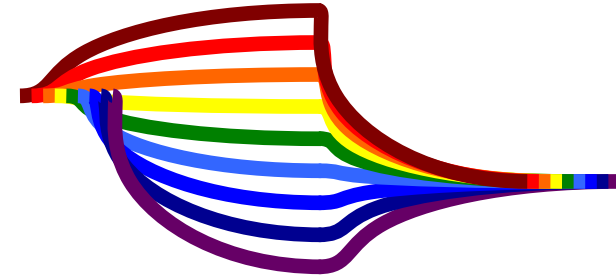
*WAVE (Wireless Access in Vehicular Environment) standards:
IEEE 802.11p standard and 1609.x standards*

IEEE 802.11{a,b,g,n} for Car-to-X communication?

- Can't be in infrastructure mode and ad hoc mode at the same time
- Switching time consuming
- Association time consuming
- No integral within-network security
- (Massively) shared spectrum ($\Rightarrow$ ISM)
- No integral QoS
- Multi-path effects reduce range and speed

IEEE 802.11p

- IEEE 802.11p
 - PHY layer mostly identical to IEEE 802.11a
 - Variant with OFDM and 16 QAM
 - Higher demands on tolerances
 - Reduction of inter symbol interference because of multi-path effects
 - Double timing parameters
 - Channel bandwidth down to 10 MHz (from 20 MHz)
 - Throughput down to 3 ... 27 Mbit/s (from 6 ... 54 Mbit/s)
 - Range up to 1000 m, speed up to 200 km/h
 - MAC layer of IEEE 802.11a plus extensions
 - Random MAC Address
 - QoS (EDCA priority access, cf. IEEE 802.11e, ...)
 - Multi-Frequency and Multi-Radio capabilities
 - New Ad Hoc mode
 - ...



Differences between IEEE 802.11p and IEEE 802.11a



Parameters	IEEE 802.11a	IEEE 802.11p half clocked mode	Changes
Bit rate (Mbit/s)	6, 9, 12, 18, 24, 36, 48, 54	3, 4.5, 6, 9, 12, 18, 24, 27	Half
Modulation mode	BPSK, QPSK, 16QAM, 64QAM	BPSK, QPSK, 16QAM, 64QAM	No change
Code rate	1/2, 2/3, 3/4	1/2, 2/3, 3/4	No change
Number of subcarriers	52	52	No change
Symbol duration	4 μ s	8 μ s	Double
Guard time	0.8 μ s	1.6 μ s	Double
FFT period	3.2 μ s	6.4 μ s	Double
Preamble duration	16 μ s	32 μ s	Double
Subcarrier spacing	0.3125 MHz	0.15625 MHz	Half



Differences between IEEE 802.11p and IEEE 802.11a

- **Narrower bandwidth (10 MHz)**
 - **Halves** the occupied frequency space compared to 20 MHz in 802.11a
 - **You speak fewer words per second but more clearly,**
 - so **the listener can catch them despite wind and echoes.**
- **Carrier spacing halved**
 - The 802.11p signal uses a carrier spacing reduced by $\frac{1}{2}$ compared to 802.11a.
- **Symbol length doubled**
 - Symbol time = $1 \div \text{carrier spacing}$.
 - Halving the spacing doubles the symbol duration (from 4 μs to 8 μs , plus guard interval), improving tolerance to multipath delay spread.

"Narrower bandwidth (10 MHz) → Carrier spacing halved → Symbol length doubled

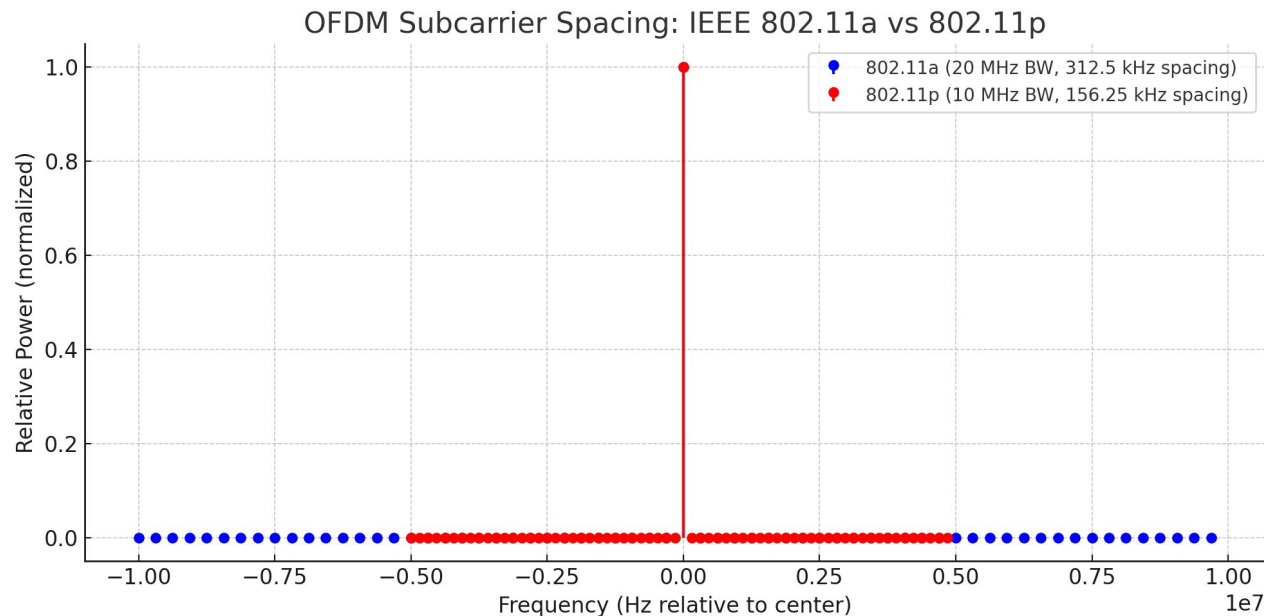
Halves data rate-->doubles symbol time = improving tolerance to multipath delay spread.

Differences between IEEE 802.11p and IEEE 802.11a

- **Carrier spacing halved**

The 802.11p signal uses a carrier spacing reduced by $\frac{1}{2}$ compared to 802.11a.

- **carrier spacing = bandwidth $\div$ number of subcarriers.**
- Keeping 52 subcarriers but halving bandwidth from 20 MHz to 10 MHz halves the spacing (from 312.5 kHz to 156.25 kHz).



Differences between IEEE 802.11p and IEEE 802.11a

Halves data rate-->doubles symbol time = improving tolerance to multipath delay spread.

- Think of it like **cars on a highway**:
- In **802.11a (blue)**, the lanes (carrier spacing) are wide apart, and cars are going very fast (short symbol time).
- That's fine on a smooth, straight road, but if there are bumps (signal reflections) or gusts of wind (Doppler shifts), you can easily lose control because everything happens so quickly.
- In **802.11p (red)**, the lanes are closer together (smaller carrier spacing) and all cars drive slower (longer symbol time).
- Because they're moving slower, they have more reaction time if a bump or gust happens —
- meaning **the system can handle reflections and frequency shifts without “crashing” the communication.**

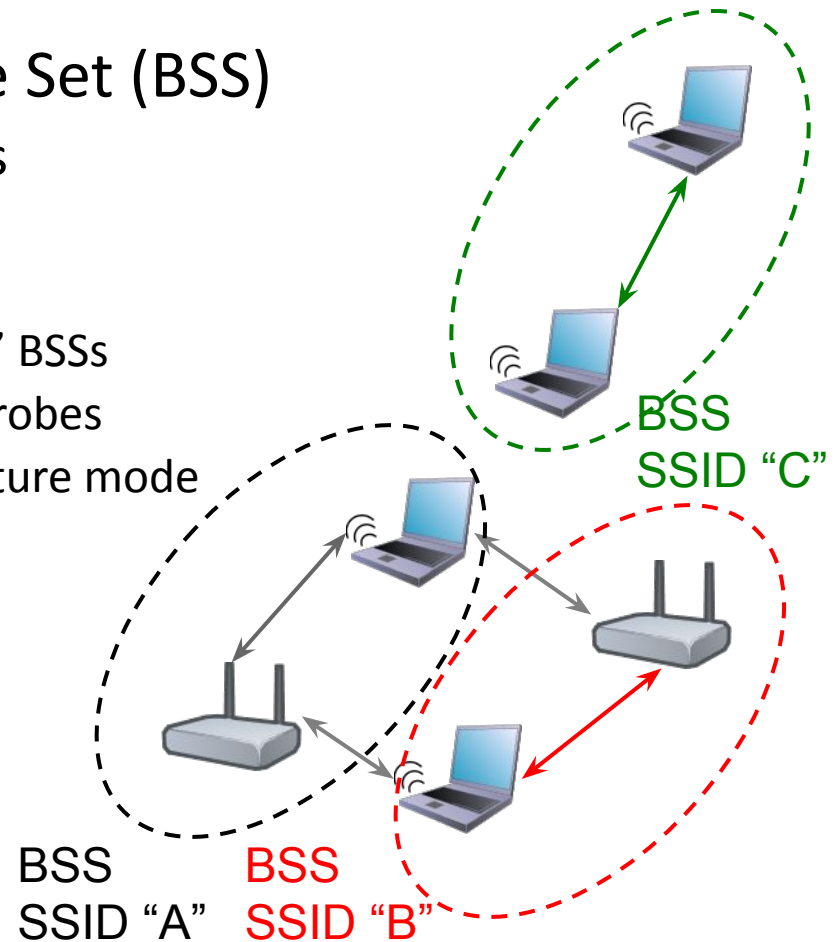
Classic IEEE 802.11

- Classic IEEE 802.11 Basic Service Set (BSS)

- Divides networks into logical units
 - Nodes belong to (exactly one) BSS
 - Packets contain BSSID
 - Nodes ignore packets from “foreign” BSSs
 - Exception: Wildcard-BSSID (-1) for probes
 - Ad hoc networks emulate infrastructure mode

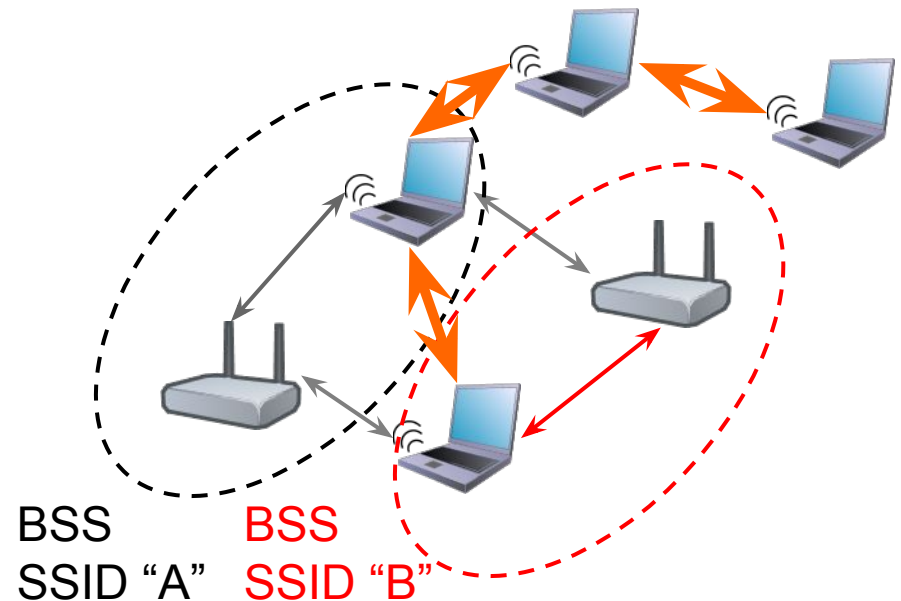
- Joining a BSS

- Access Point sends beacon
- Authentication dialogue
- Association dialogue
- Node has joined BSS



IEEE 802.11p

- New: 802.11 WAVE Mode
 - Default mode of nodes in WAVE
 - Nodes may always use Wildcard BSS in packets
 - Nodes will always receive Wildcard BSS packets
 - May join BSS and still use Wildcard BSS

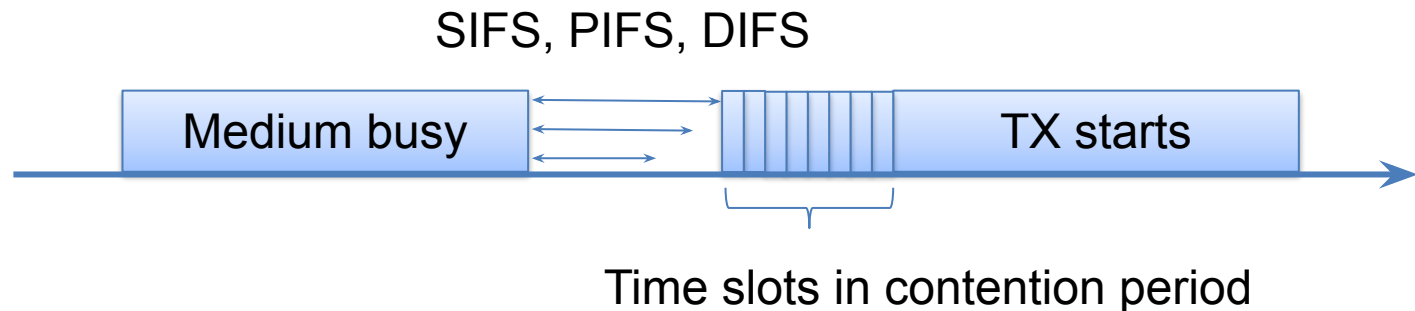


IEEE 802.11p

- New: 802.11 WAVE BSS
 - No strict separation between Host and Access Point (AP)
 - Instead, loose classification according to:
 - Equipment: Roadside Unit (RSU) / On-Board Unit (OBU)
 - Role in data exchange: Provider (offers service) / User (uses service)
 - No technical difference between Provider and User —no hardware difference.
 - Provider sends On-Demand Beacon
 - Analogous to standard 802.11-Beacon
 - Beacon contains all information and parameters needed to join
 - User configures lower layers accordingly
 - Starts using provided service
 - No additional exchange of data needed
 - BSS membership now only implied
 - There's no formal “join” process.
 - Once a beacon is received and used, the User is considered part of the BSS.
 - BSS continues to exist even after provider leaves

IEEE 802.11 DCF

- IEEE 802.11 Distributed Coordination Function (DCF)
 - aka “Contention Period”



- Priority access via SIFS (ACK, CTS, ...) and DIFS (payload)
- Wait until medium has been free for duration of DIFS
- If medium busy, wait until idle, then wait DIFS plus random backoff time

IEEE 802.11 DCF

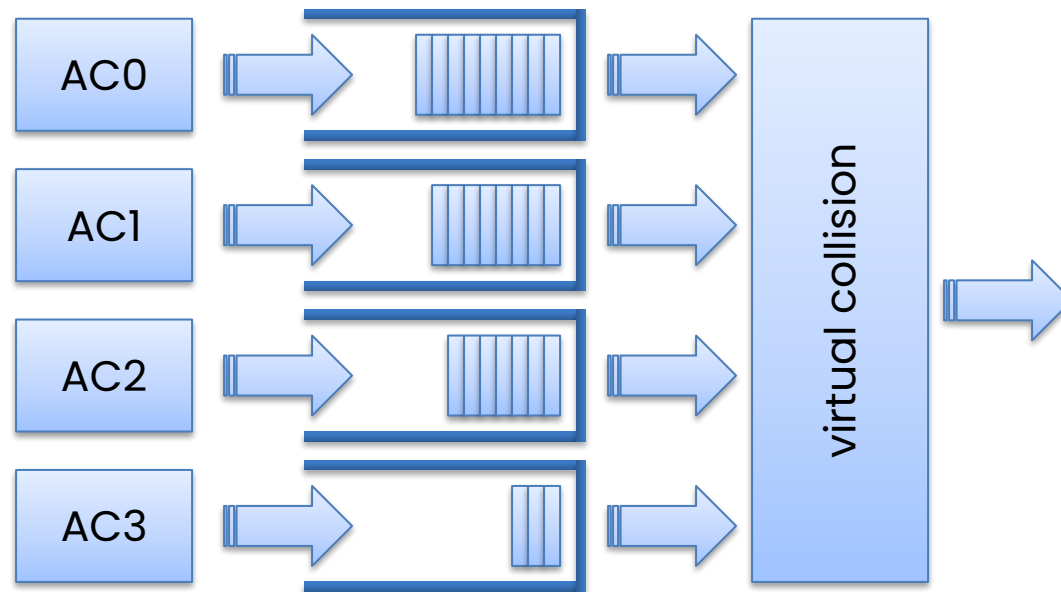
- IEEE 802.11 Distributed Coordination Function (DCF)
 - Backoff if
 - a) Node is ready to send and channel becomes busy
 - b) A higher priority queue ($\Rightarrow$ next slides) becomes ready to send
 - c) Unicast transmission failed (no ACK)
 - d) Transmission completed successfully
 - Backoff: Random slot count from interval $[0, CW]$
 - Decrement by one after channel was idle for one slot (only in contention period)
 - In cases b) and c), double CW (but no larger than $CW_{\max}$)
 - In case d), set CW to $CW_{\min}$

IEEE 802.11p

- QoS in 802.11p (HCF)
 - cf. IEEE 802.11e EDCA
 - DIFS $\Rightarrow$ AIFS (Arbitration Inter-Frame Space)
 - DCF $\Rightarrow$ EDCA (Enhanced Distributed Channel Access)
 - Classify user data into 4 ACs (Access Categories)
 - AC0 (lowest priority)
 - ...
 - AC3 (highest priority)
 - Each ACs has different...
 - $CW_{\min}$, $CW_{\max}$, AIFS, TXOP limit (max. continuous transmissions)
- Management data e.g, WAVE Service Advertisement (WSA), Timing synchronization) uses DIFS (not AIFS)

IEEE 802.11p

- QoS in 802.11p (HCF)
 - Map 8 user priorities (Min:0 Max:7) $\Rightarrow$ 4 access categories $\Rightarrow$ 4 queues
 - Queues compete independently for medium access



IEEE 802.11p

■ QoS in 802.11p (HCF)

- IEEE 802.11e defines 8 possible priorities (0–7),
- but EDCA only supports 4 queues in hardware:
 - AC_VO (Voice)
 - AC_VI (Video)
 - AC_BE (Best Effort)
 - AC_BK (Background)

• **AC_VO** → Priority (P):6-7 # Used for safety-critical messages (e.g., BSM) → (shortest AIFS, smallest contention window.)

• **AC_VI** → Priority (P):4-5 # Non-critical but time-sensitive infotainment.

• **AC_BE** → Priority (P):1-3 # Default for general data.

• **AC_BK** → Priority (P):0 # Low-priority background updates.

IEEE 802.11p

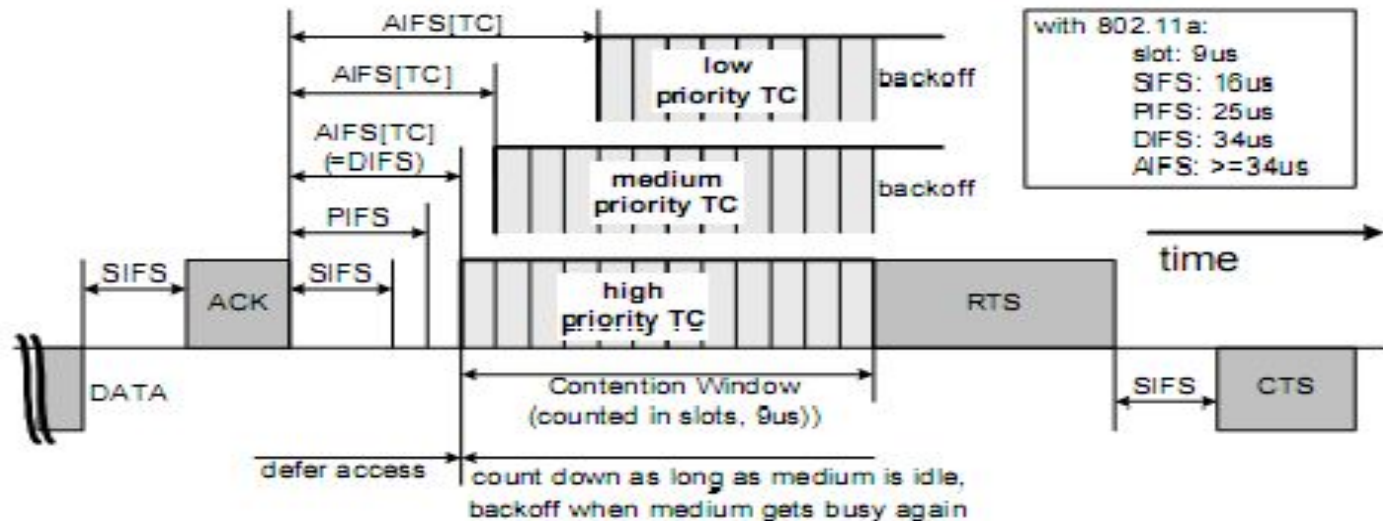
- QoS in 802.11p (HCF)
 - Parameterization

Parameter	Value
SlotTime	13μs
SIFS	32μs
CW _{min}	15
CW _{max}	1023
Bandwidth	3 .. 27 mbit/s

Parameter	AC_BK	AC_BE	AC_VI	AC_VO
CW _{min}	CW _{min}	CW _{min}	$[(CW_{min}+1)/2]-1$	$[(CW_{min}+1)/4]-1$
CW _{max}	CW _{max}	CW _{max}	CW _{min}	$[(CW_{min}+1)/2]-1$
AIFS _n	9	6	3	2

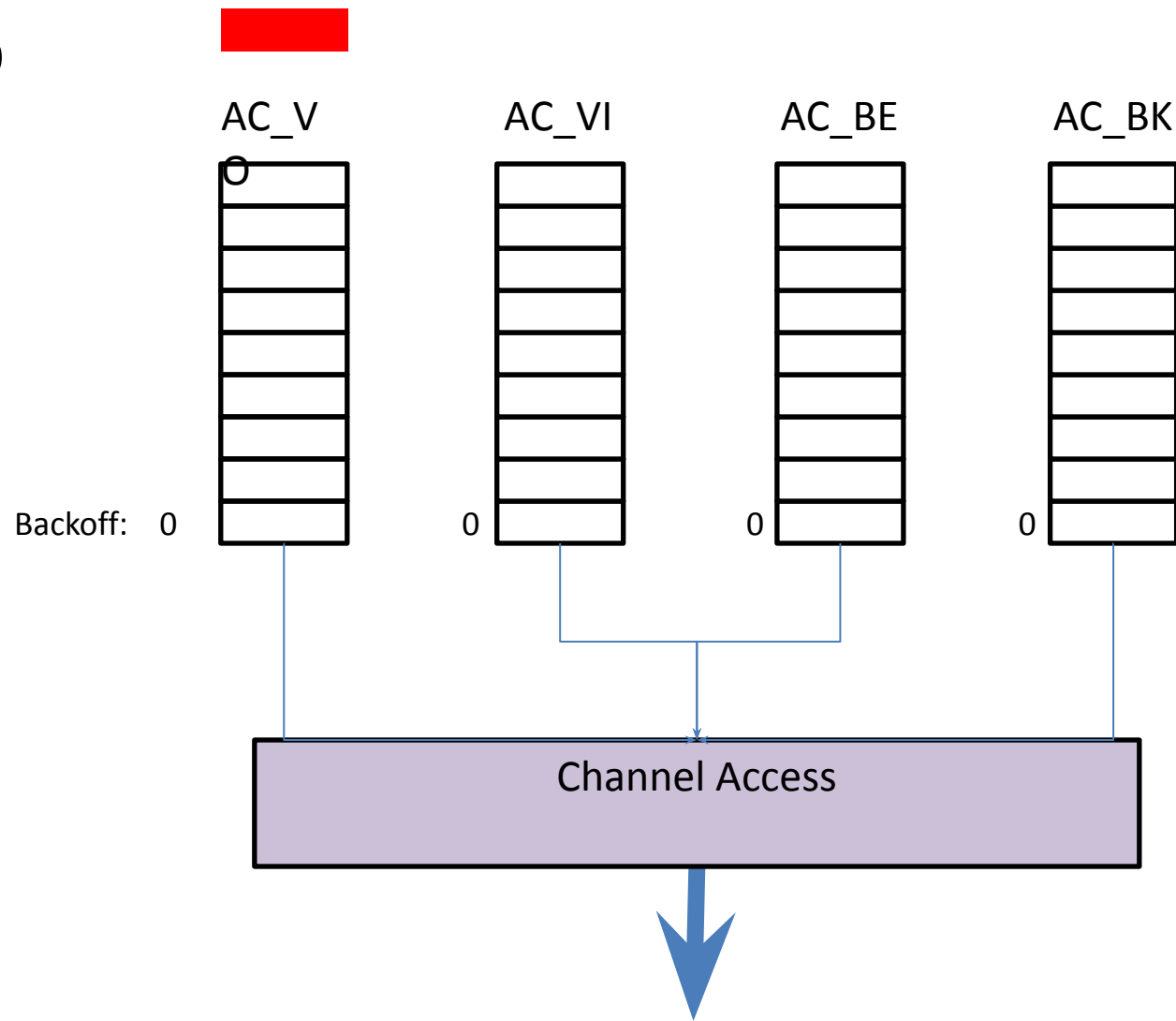
Classical IEEE 802.11

QoS in 802.11p (HCF) Parameterization

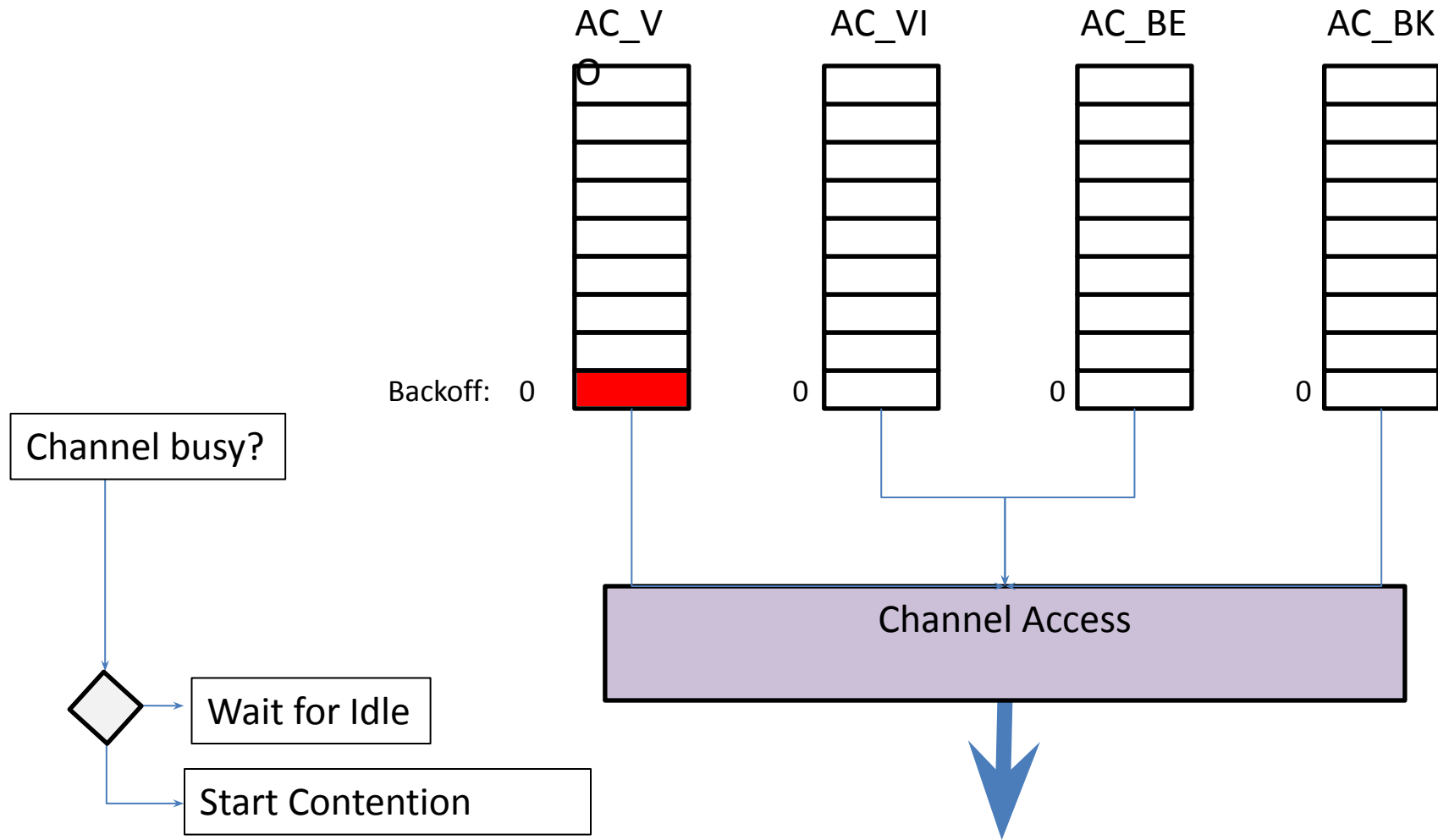


- After detecting the medium being idle for a duration of AIFS[AC] (Arbitration Interframe Space), backoff entity start down-counting. AIFS is calculated as follow:
 - $AIFS[AC] = SIFS + AIFSN[AC] \cdot aSlotTime \quad \geq 2$
- Min and Max of the contention window are the other parameters dependent on AC.

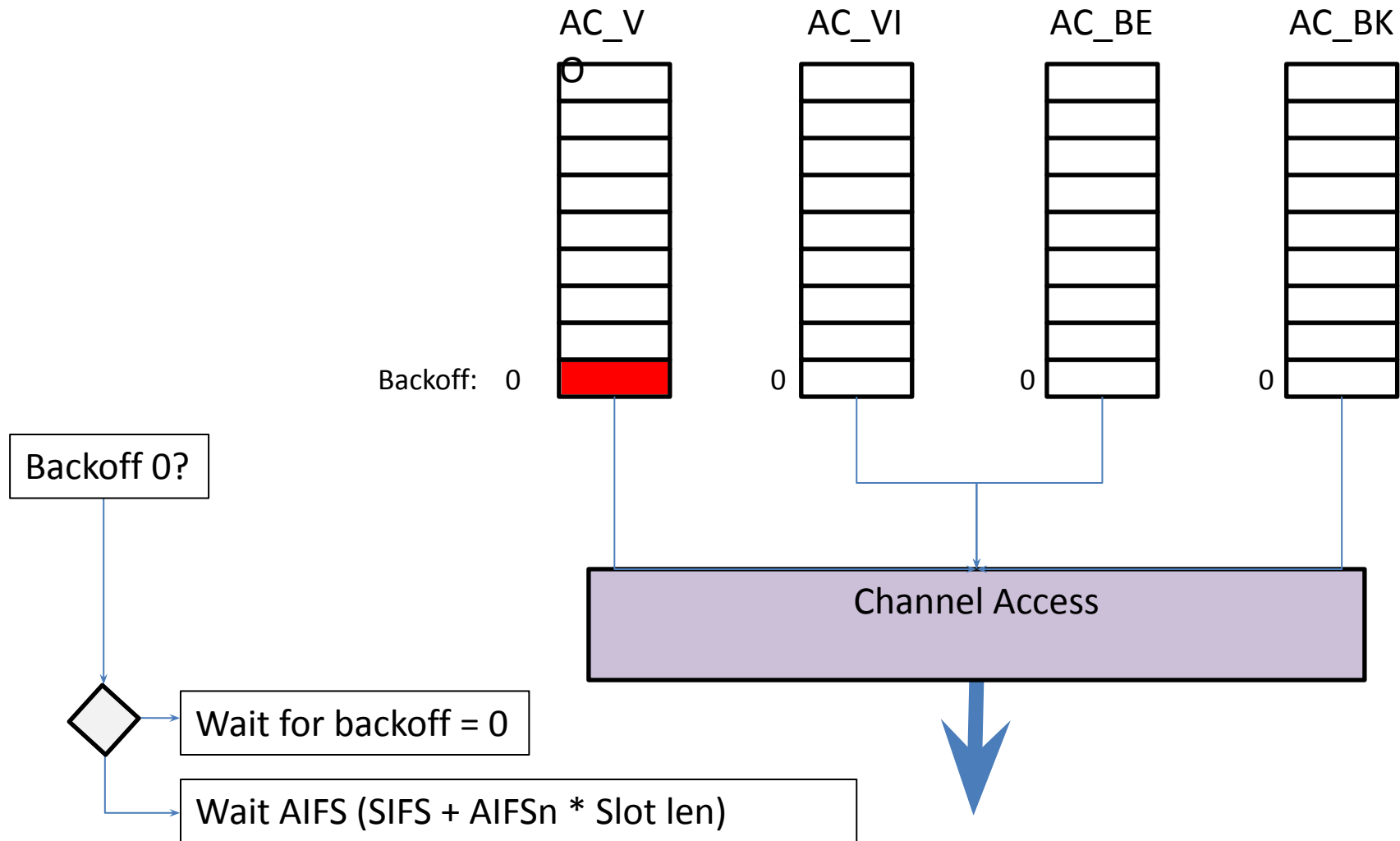
IEEE 802.11p



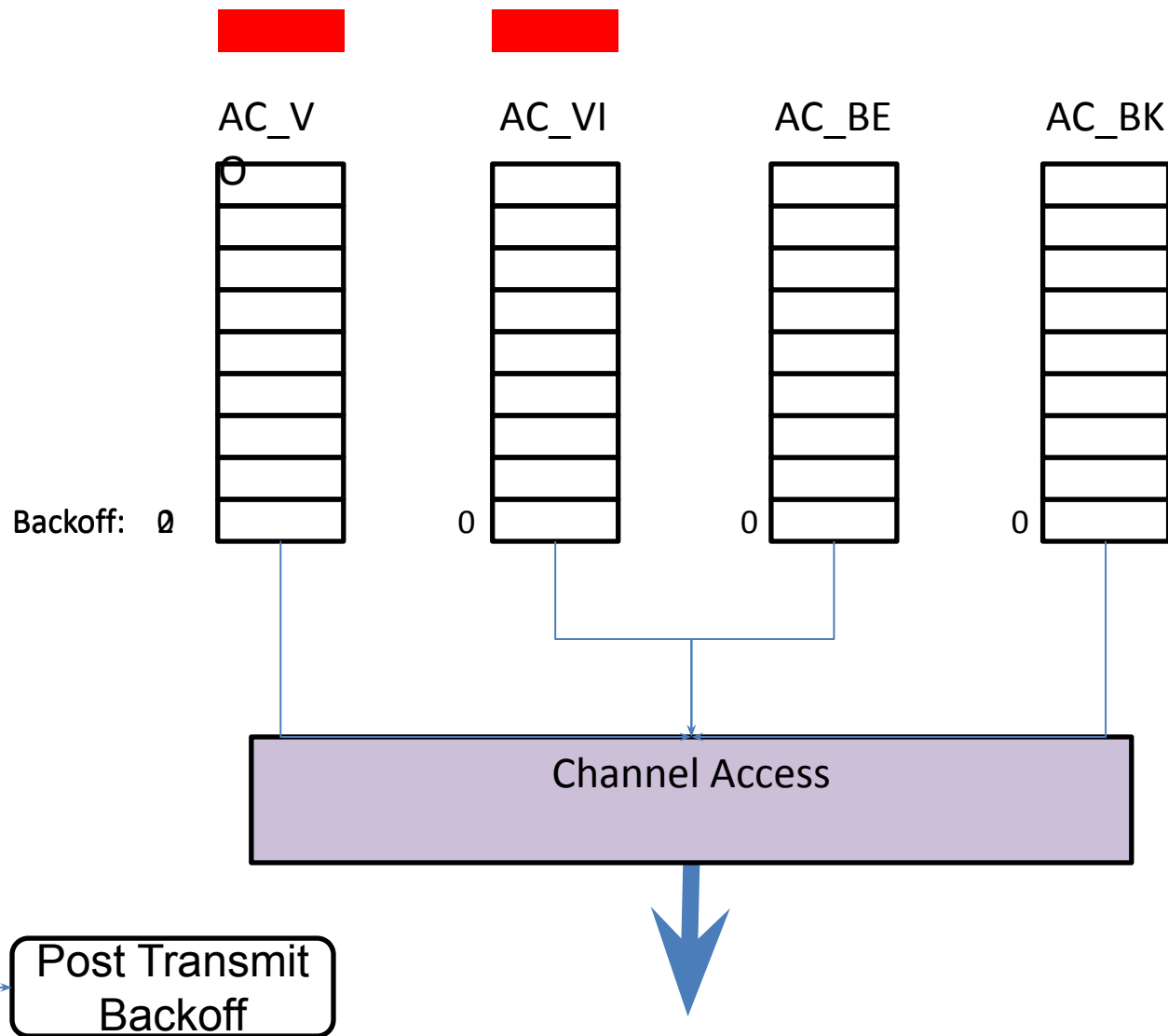
IEEE 802.11p



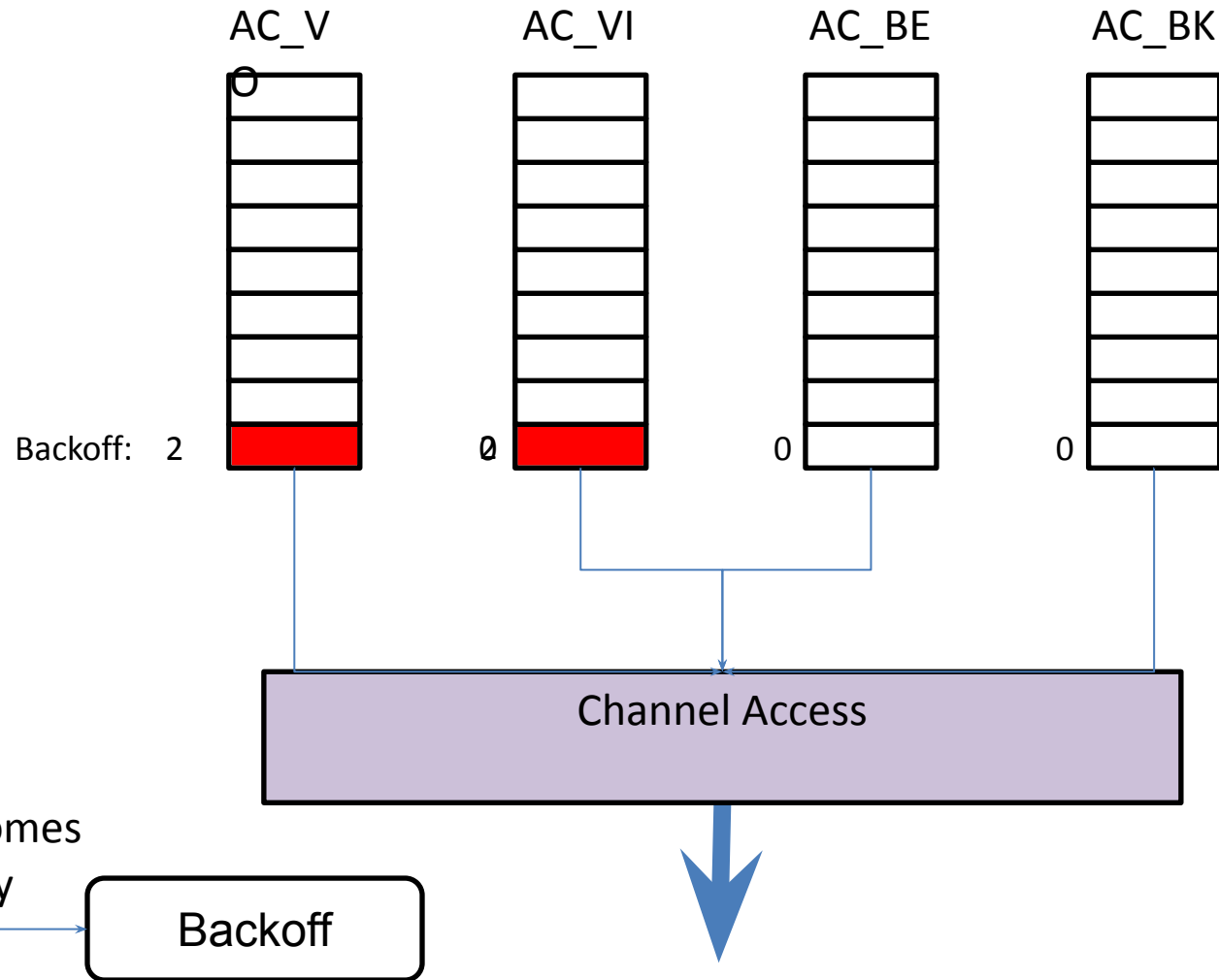
IEEE 802.11p



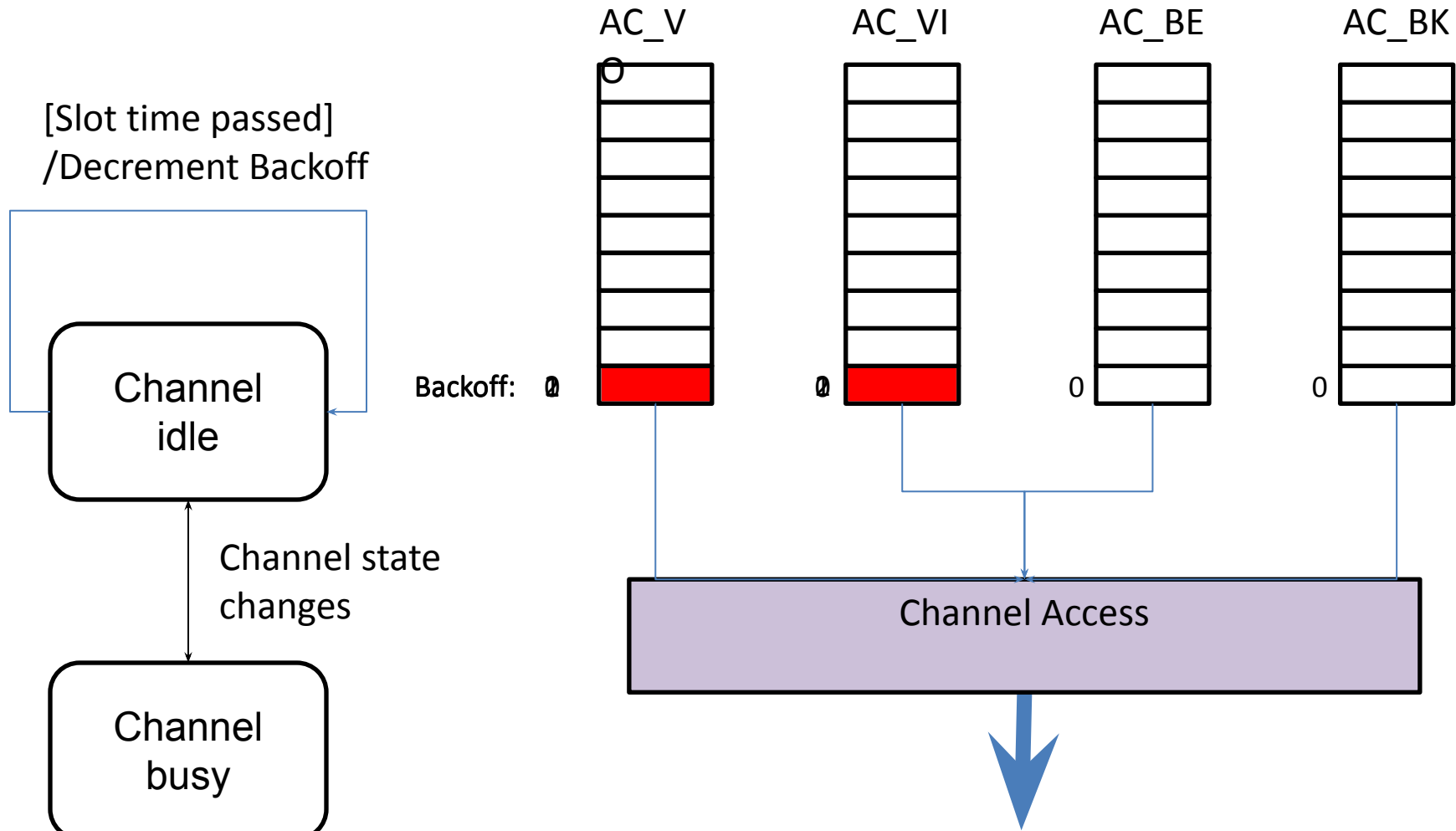
IEEE 802.11p



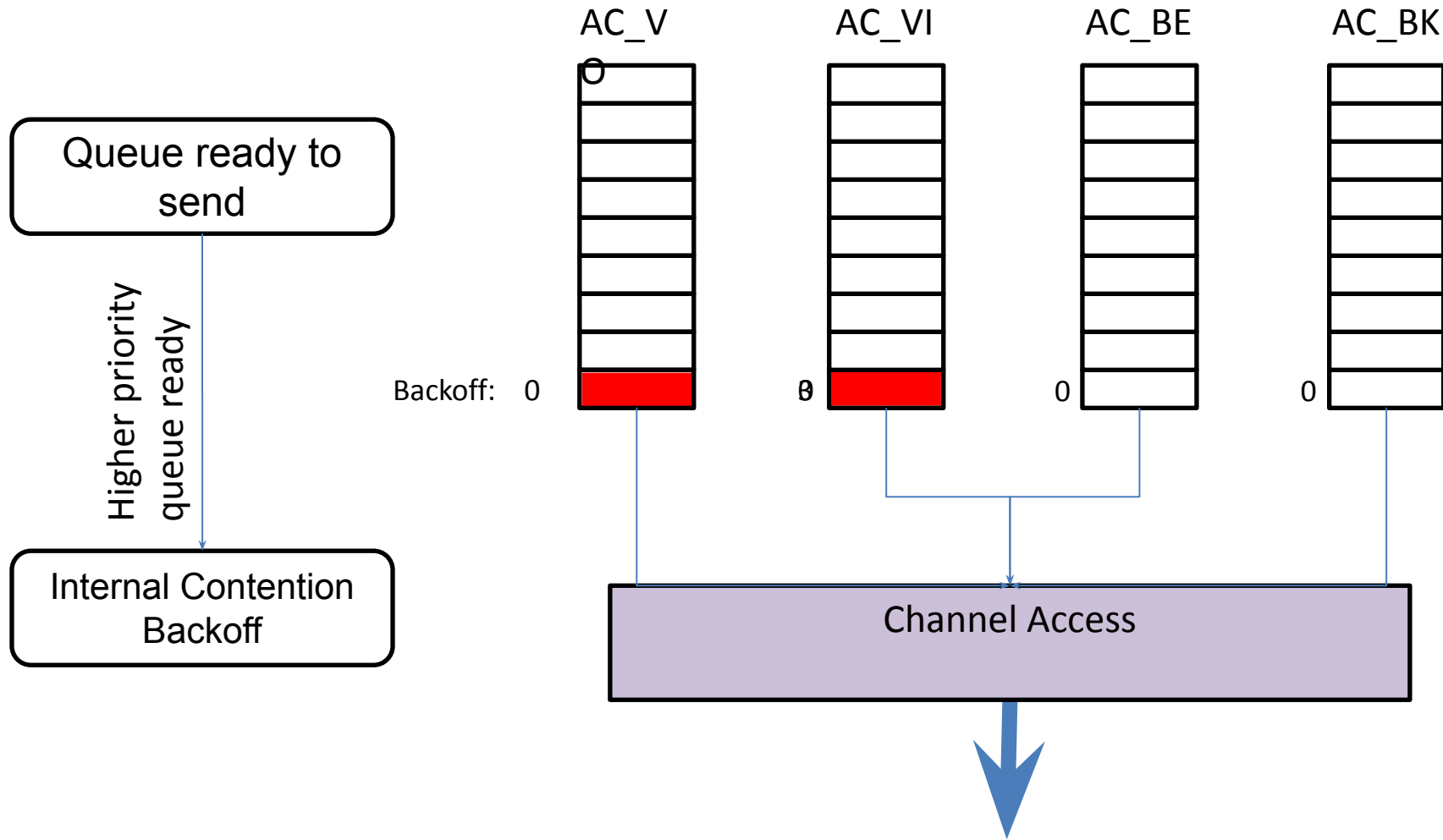
IEEE 802.11p



IEEE 802.11p



IEEE 802.11p



IEEE 802.11p

- QoS in WAVE
 - mean waiting time for channel access, given sample configuration (and TXOP Limit=0)
 - ⇒ single packet
 - when channel idle:
 - when channel busy:

AC	CW _{min}	CW _{max}	AIFS	TXOP	t _w (in μs)
0	15	1023	9	0	264
1	7	15	6	0	152
2	3	7	3	0	72
3	3	7	2	0	56

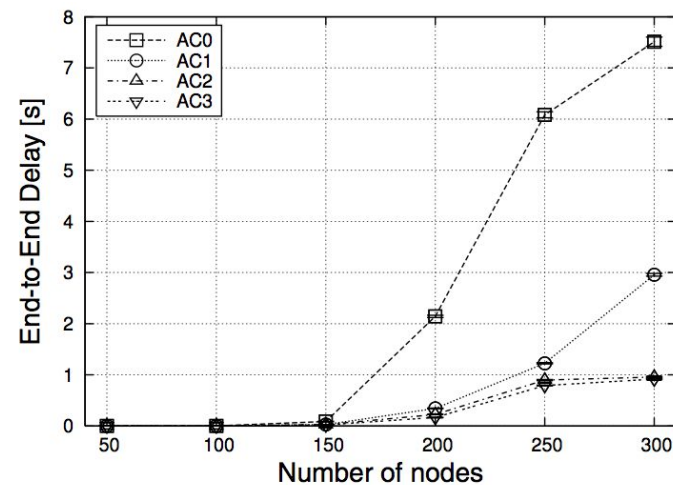


Figure Source: Eichler, S., "Performance evaluation of the IEEE 802.11p WAVE communication standard," Proceedings of 66th IEEE Vehicular Technology Conference (VTC2007-Fall), Baltimore, USA, October 2007, pp. 2199-2203

UMTS/LTE vs. 802.11p

- Pros of UMTS/LTE

- + Easy provision of centralized services
- + Quick dissemination of information in whole network
- + Pre-deployed infrastructure
- + Easy migration to (and integration into) smartphones

- Cons of UMTS/LTE

- High short range latencies (might be too high for safety)
- Network needs further upgrades (smaller cells, multicast service)
- High dependence on network operator
- High load in core network, even for local communication

UMTS/LTE vs. IEEE 802.11p

- Pros of 802.11p/Ad hoc
 - + Smallest possible latency
 - + Can sustain operation without network operator / provider
 - + Network load highly localized
 - + Better privacy (⇒ later slides)
- Cons of 802.11p/Ad hoc
 - Needs gateway for provision of central services (e.g., RSU)
 - No pre-deployed hardware, and hardware is still expensive
- The solution?
 - hybrid systems:
deploy both technologies to vehicles and road,
decide depending on application and infrastructure availability

Higher Layer Standards for IEEE 802.11p

- Channel management
 - Dedicated frequency band at 5.9 GHz allocated to WAVE
 - Exclusive for V2V und V2I communication
 - No license cost, but strict rules
 - 1999: FCC reserves 7 channels of 10 MHz (“U.S. DSRC”)
 - 2 reserved channels, 1+4 channels for applications
 - ETSI Europe reserves 5 channels of 10 MHz

U.S. allocation	...	Critical Safety of Life	SCH	SCH	Control Channel (CCH)	SCH	SCH	Hi-Power Public Safety	...
European allocation		SCH	SCH	SCH	SCH	CCH	SCH	SCH	
IEEE Channel		172	174	176	178	180	182	184	
Center frequency		5.860 GHz	5.870 GHz	5.880 GHz	5.890 GHz	5.900 GHz	5.910 GHz	5.920 GHz	

[1] ETSI ES 202 663 V1.1.0 (2010-01) : Intelligent Transport Systems (ITS); European profile standard for the physical and medium access control layer of Intelligent Transport Systems operating in the 5 GHz frequency band

DSRC:: Channel Allocation (U.S. allocation)

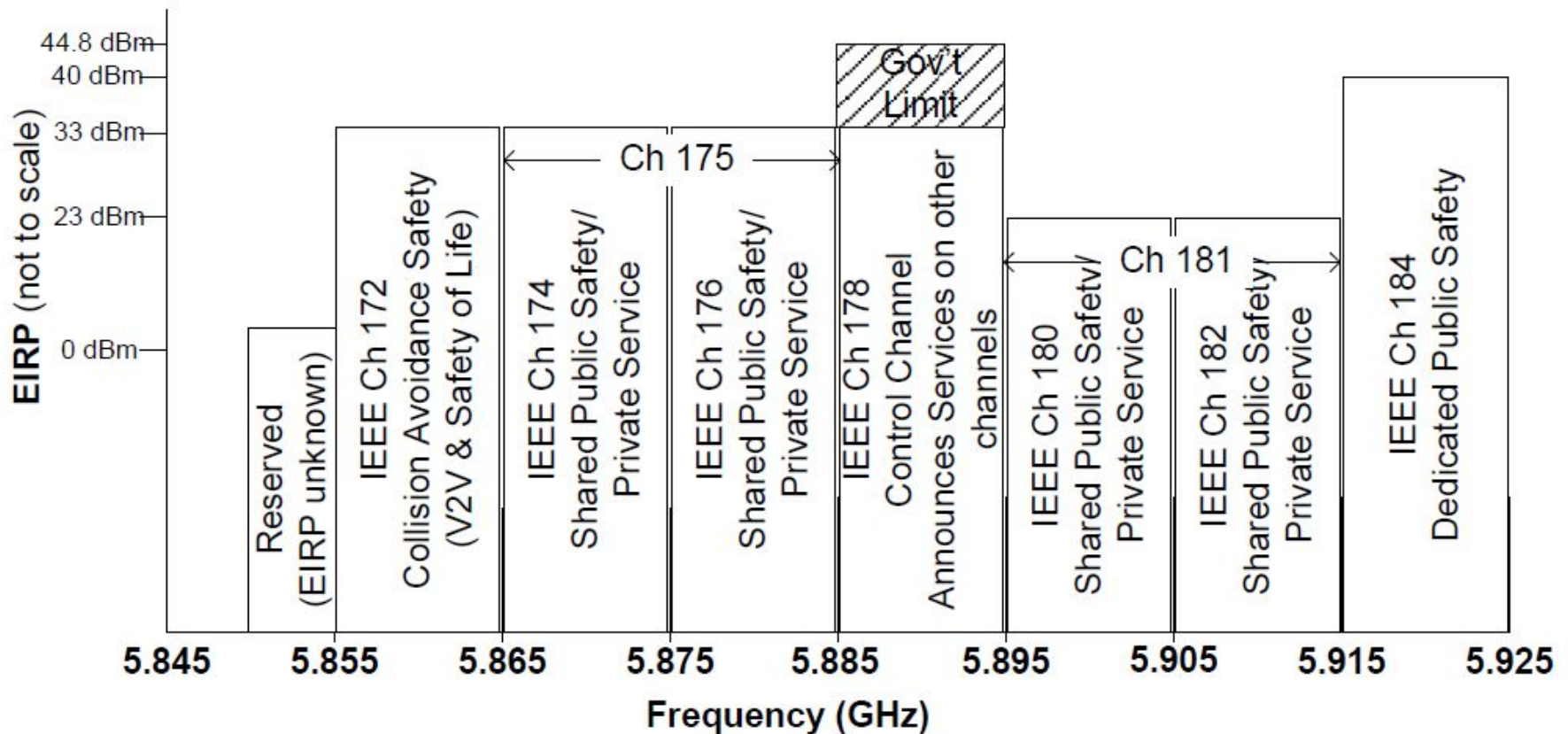
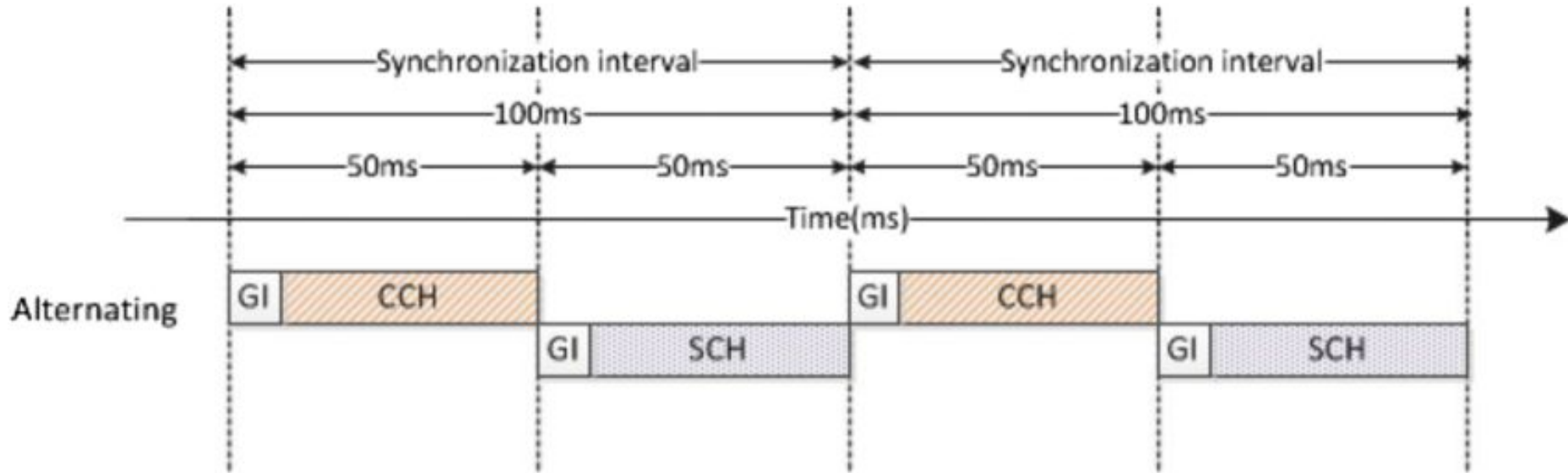


Figure 3-1: US ITS/DSRC channel plan

DSRC:: Channel Switching Timing



Safety Applications :

Primary Message Type: **BSM (Basic Safety Message)** – Periodically broadcast vehicle status (position, speed, heading).

Channel Used: **Control Channel (CCH)** or **Safety Channel (SfCH)**

Message Type and Channel Used

Safety Applications :

- **Primary Message Type:** **BSM (Basic Safety Message)** – Periodically broadcast vehicle status (position, speed, heading).
- **Channel Used:** **Control Channel (CCH)** or **Safety Channel (SfCH)**

Non-Safety Applications :

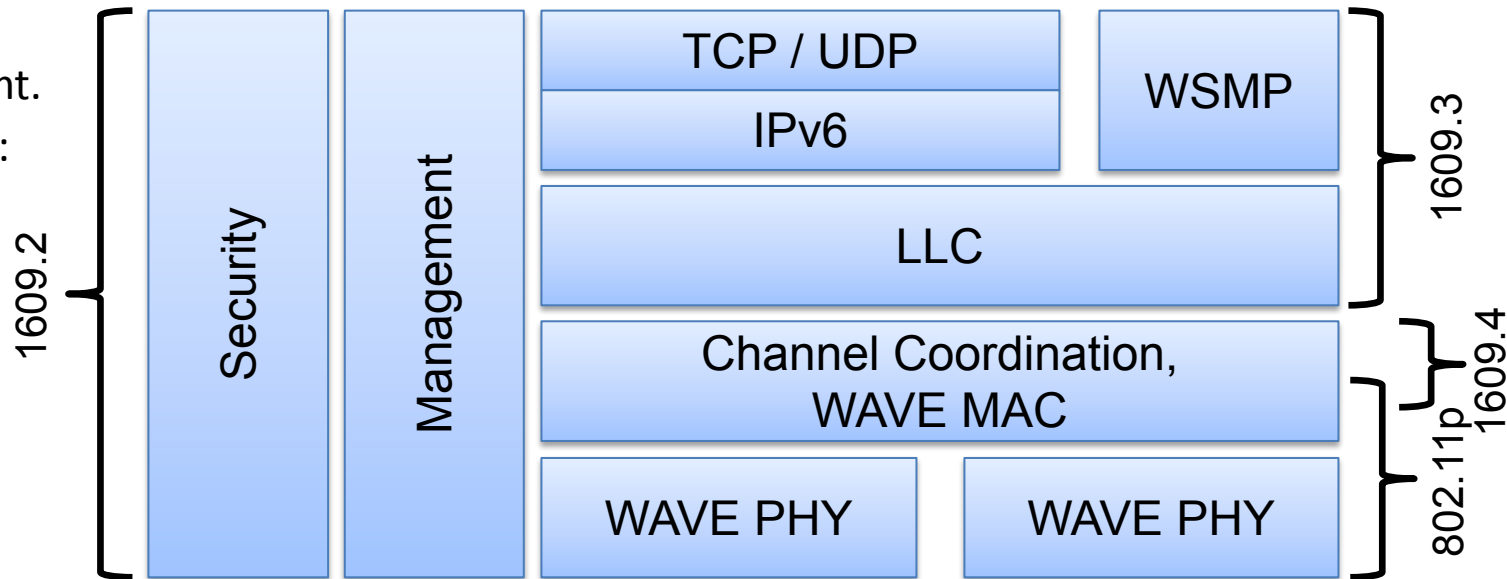
- **Primary Message Types:**
 - **WSA (WAVE Service Advertisement)** – Broadcasts the availability of a service on a specific channel.
 - **WSM (WAVE Short Message)** – Payload message sent after WSA received/accepted.
- **Channel Used:** **Service Channels (SCHs)** (e.g., 5.860 GHz–5.880 GHz, 5.910 GHz–5.920 GHz)

Higher Layer Standards for IEEE 802.11p

- Need for higher layer standards
 - Unified message format
 - Unified interfaces to application layer
- U.S.
 - IEEE 1609.*
 - WAVE („Wireless Access in Vehicular Environments“)
- Europe
 - ETSI
 - ITS G5 (“Intelligent Transportation Systems“)

IEEE 1609.* upper layers (building on IEEE 802.11p)

- IEEE 1609.2:
Security
- IEEE 1609.3:
Network
services
- IEEE 1609.4:
Channel mgmt.
- IEEE 1609.11:
Application
“electronic
payment”



[1] Jiang, D. and Delgrossi, L., "IEEE 802.11p: Towards an international standard for wireless access in vehicular environments," Proceedings of 67th IEEE Vehicular Technology Conference (VTC2008-Spring), Marina Bay, Singapore, May 2008

[2] Uzcátegui, Roberto A. and Acosta-Marum, Guillermo, "WAVE: A Tutorial," IEEE Communications Magazine, vol. 47 (5), pp. 126-133, May 2009

IEEE 1609

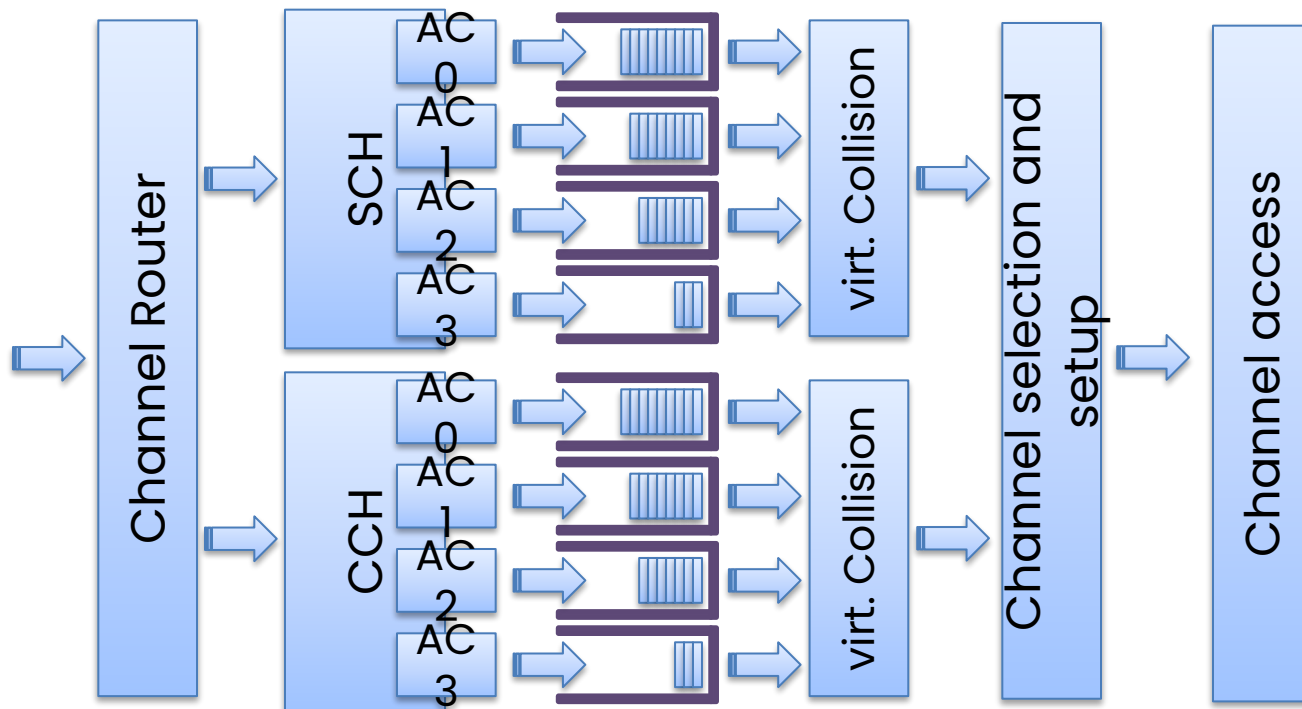
- Channel management
 - WAVE allows for both single radio devices & multi radio devices
 - Dedicated Control Channel (CCH) for mgmt and safety messages
 - ⇒ single radio devices need to periodically listen to CCH
 - Time slots
 - Synchronization envisioned via GPS receiver clock
 - Standard value: 100ms sync interval (with 50ms on CCH)
 - Short guard interval at start of time slot
 - During guard, medium is considered busy (⇒ backoff)



[1] IEEE Vehicular Technology Society, "IEEE 1609.4 (Multi-channel Operation)," IEEE Std, November, 2006

IEEE 1609

- Packet transmission
 - Sort into AC queue, based on WSMP (or IPv6) EtherType field, destination channel, and user priority
 - Switch to desired channel, setup PHY power and data rate
 - Start medium access

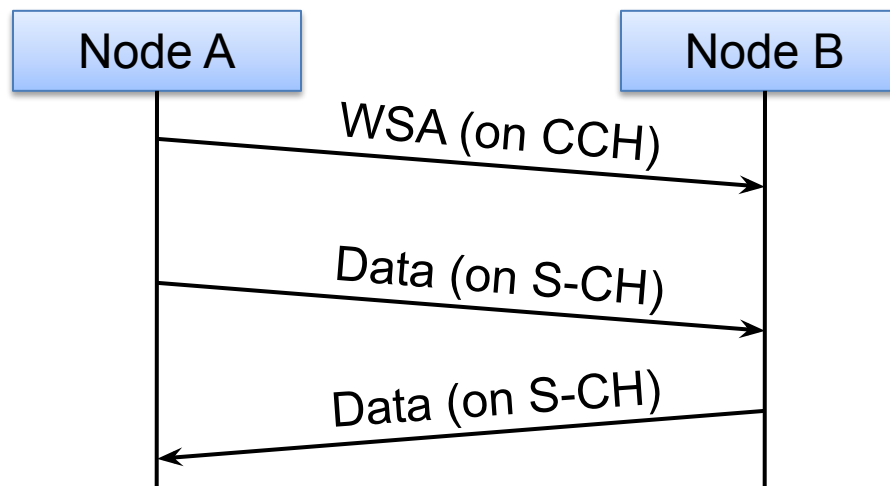


IEEE 1609

- Channel management
 - Control Channel (CCH):
 - Default channel upon initialization
 - BSM(Basic Safety Message), WAVE service advertisements (WSA)
 - Channel parameters take fixed values
 - Service Channel (SCH):
 - Only after joining WAVE BSS
 - WAVE short messages (WSM),
IP data traffic (IPv6)
 - Channel parameters can be changed as needed

IEEE 1609

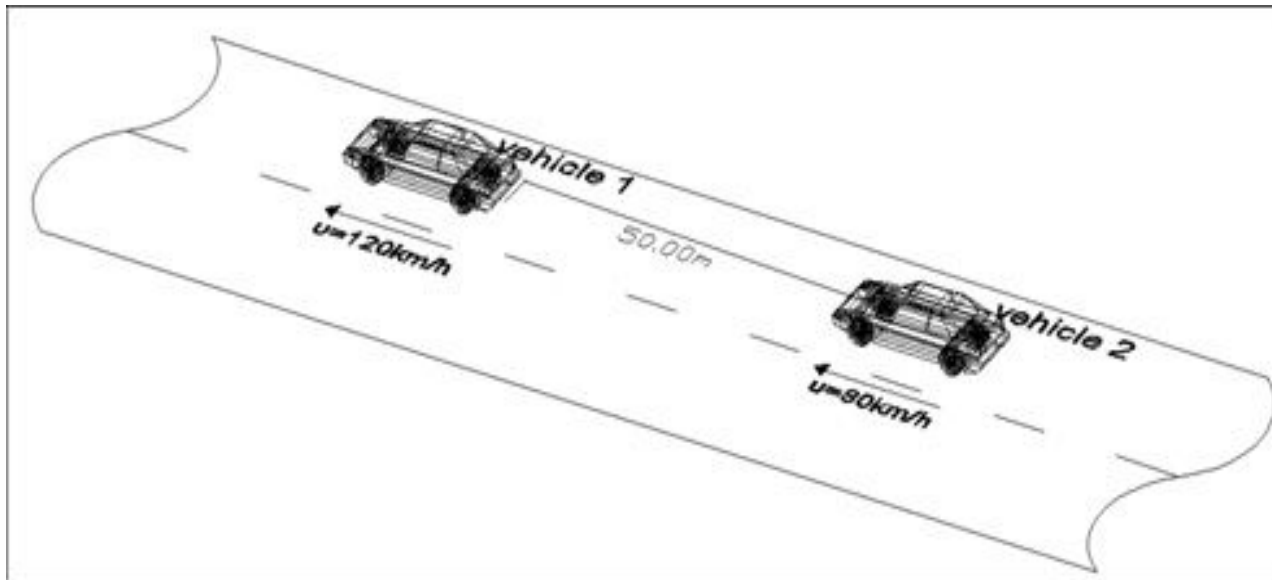
- WAVE service advertisement (WSA)
 - Broadcast on Control Channel (CCH)
 - Identifies WAVE BSSs on Service Channels (SCHs)
 - Can be sent at arbitrary times, by arbitrary nodes
 - Only possibility to make others aware of data being sent on SCHs, as well as the required channel parameters to decode them



Loss Model

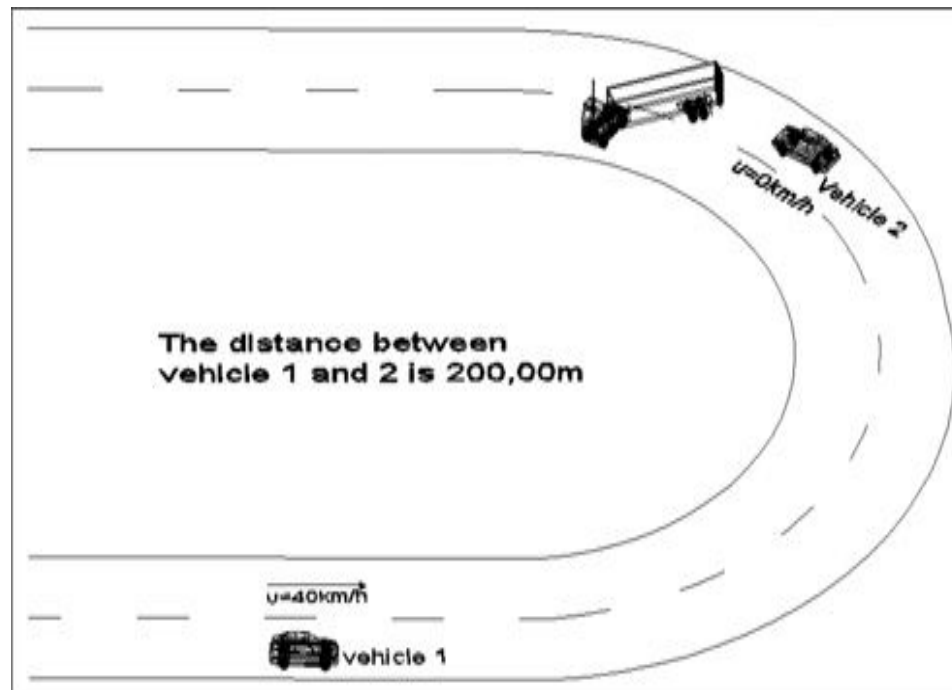
Loss Model::Line Of Sight

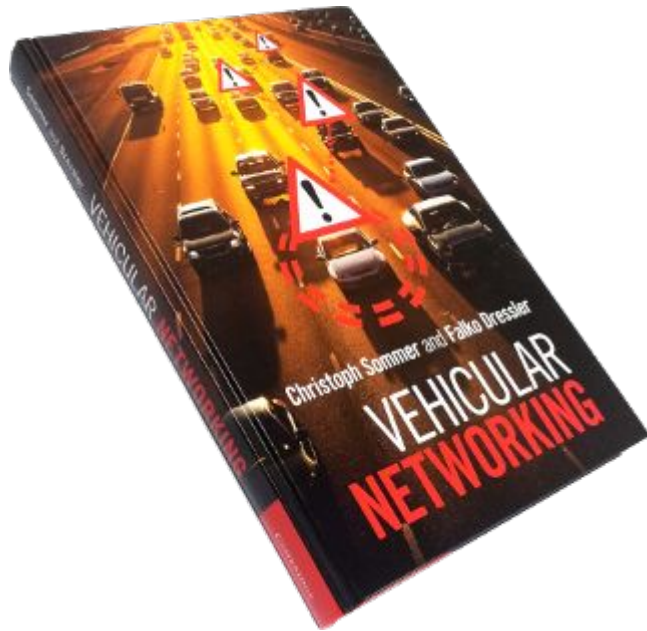
- The term LOS path is referred the situation where we have visual line of sight contact between the transmitting antenna and the receiving antenna.
- The **Ricean distribution** can be used for modeling small-scale fading



Loss Model:: Non-Line Of Sight

- For the N-LOS path it is found that the amplitude of the received signal due to fading obeys **Rayleigh fading distribution** and the delay spreads were several times greater than that for the LOS.





Vehicular Networking
